

# Chapter 6



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The train produces a moving load that must be considered using influence lines when designing the girders of this bridge.

# Influence Lines for Statically Determinate Structures

Influence lines have important application for the design of structures that resist large live loads. In this chapter we will discuss how to draw the influence line for a statically determinate structure. The theory is applied to structures subjected to a distributed load or a series of concentrated forces, and specific applications to floor girders and bridge trusses are given. The determination of the absolute maximum live shear and moment in a member is discussed at the end of the chapter.

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## 6.1 Influence Lines

In the previous chapters we developed techniques for analyzing the forces in structural members due to *dead* or *fixed loads*. It was shown that the *shear* and *moment diagrams* represent the most descriptive methods for displaying the variation of these loads in a member. If a structure is subjected to a *live* or *moving load*, however, the variation of the shear and bending moment in the member is best described using the ***influence line***. An influence line represents the variation of either the reaction, shear, or moment, at a *specific point* in a member as a concentrated force moves over the member. Once this line is constructed, one can tell at a glance where the moving load should be placed on the structure so that it creates the greatest influence at the specified point. Furthermore, the magnitude of the associated reaction, shear, or moment at the point can then be calculated from the ordinates of the influence-line diagram. For these reasons, influence lines play an important part in the design of bridges, industrial crane rails, conveyors, and other structures where loads move across their span.

Although the procedure for constructing an influence line is rather basic, one should clearly be aware of the *difference* between constructing an influence line and constructing a shear or moment diagram. Influence lines represent the effect of a *moving load* only at a *specified point* on a member, whereas shear and moment diagrams represent the effect of *fixed loads* at *all points* along the axis of the member.

## Procedure for Analysis

Either of the following two procedures can be used to construct the influence line at a specific point  $P$  in a member for any function (reaction, shear, or moment). For both of these procedures we will choose the moving force to have a *dimensionless magnitude of unity*.\*

### Tabulate Values

- Place a unit load at various locations,  $x$ , along the member, and at *each* location use statics to determine the value of the function (reaction, shear, or moment) at the specified point.
- If the influence line for a vertical force *reaction* at a point on a beam is to be constructed, consider the reaction to be *positive* at the point when it acts *upward* on the beam.
- If a shear or moment influence line is to be drawn for a point, take the shear or moment at the point as positive according to the same sign convention used for drawing shear and moment diagrams. (See Fig. 4–1.)
- All statically determinate beams will have influence lines that consist of straight line segments. After some practice one should be able to minimize computations and locate the unit load *only* at points representing the *end points* of each line segment.
- To avoid errors, it is recommended that one first construct a table, listing “unit load at  $x$ ” versus the corresponding value of the function calculated at the specific point; that is, “reaction  $R$ ,” “shear  $V$ ,” or “moment  $M$ .” Once the load has been placed at various points along the span of the member, the tabulated values can be plotted and the influence-line segments constructed.

### Influence-Line Equations

- The influence line can also be constructed by placing the unit load at a *variable* position  $x$  on the member and then computing the value of  $R$ ,  $V$ , or  $M$  at the point as a function of  $x$ . In this manner, the equations of the various line segments composing the influence line can be determined and plotted.

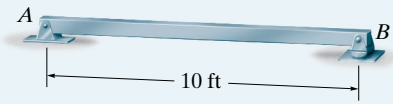
\*The reason for this choice will be explained in Sec. 6.2.

**EXAMPLE 6.1**

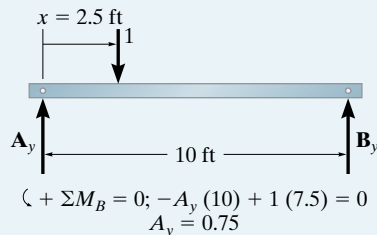
Construct the influence line for the vertical reaction at  $A$  of the beam in Fig. 6-1a.

**SOLUTION**

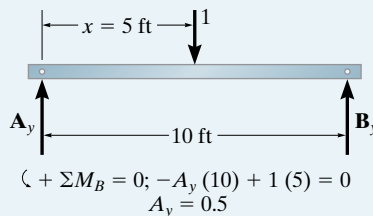
**Tabulate Values.** A unit load is placed on the beam at each selected point  $x$  and the value of  $A_y$  is calculated by summing moments about  $B$ . For example, when  $x = 2.5$  ft and  $x = 5$  ft, see Figs. 6-1b and 6-1c, respectively. The results for  $A_y$  are entered in the table, Fig. 6-1d. A plot of these values yields the influence line for the reaction at  $A$ , Fig. 6-1e. This line or diagram gives the reaction at  $A$  as the unit load moves from one position to the next along the beam.



(a)

**Fig. 6-1**

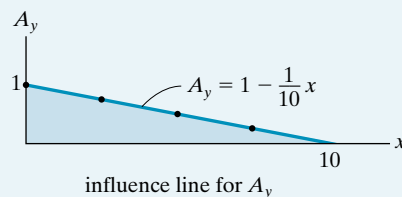
(b)



(c)

| $x$ | $A_y$ |
|-----|-------|
| 0   | 1     |
| 2.5 | 0.75  |
| 5   | 0.5   |
| 7.5 | 0.25  |
| 10  | 0     |

(d)



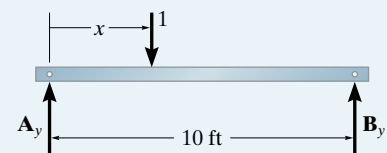
(e)

**Influence-Line Equation.** Rather than placing the unit load at *specific points* on the beam and plotting the values of  $A_y$  for each point, we can place the unit load a variable distance  $x$  from  $A$ , Fig. 6-1f. The reaction  $A_y$  as a function of  $x$  can be determined from

$$\zeta + \Sigma M_B = 0; -A_y(10) + (10 - x)(1) = 0$$

$$A_y = 1 - \frac{1}{10}x$$

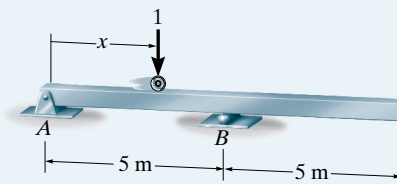
This is the equation of the influence line, plotted in Fig. 6-1e.



(f)

**EXAMPLE 6.2**

Construct the influence line for the vertical reaction at  $B$  of the beam in Fig. 6-2a.



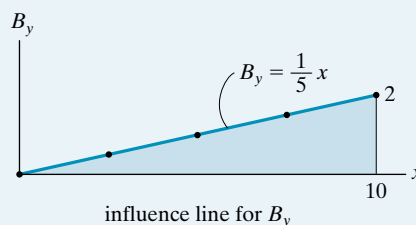
(a)

**Fig. 6-2****SOLUTION**

**Tabulate Values.** Using statics, verify that the values for the reaction  $B_y$  listed in the table, Fig. 6-2b, are correctly computed for each position  $x$  of the unit load. A plot of the values yields the influence line in Fig. 6-2c.

| $x$ | $B_y$ |
|-----|-------|
| 0   | 0     |
| 2.5 | 0.5   |
| 5   | 1     |
| 7.5 | 1.5   |
| 10  | 2     |

(b)



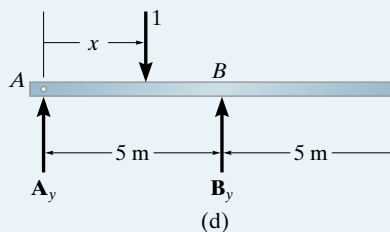
(c)

**Influence-Line Equation.** We must determine the reaction  $B_y$  when the unit load is placed a distance  $x$  from  $A$ , Fig. 6-2d. Applying the moment equation about  $A$ ,

$$\zeta + \sum M_A = 0; \quad B_y(5) - 1(x) = 0$$

$$B_y = \frac{1}{5}x$$

This is plotted in Fig. 6-2c.



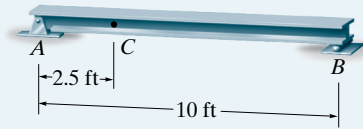
(d)

**EXAMPLE 6.3**

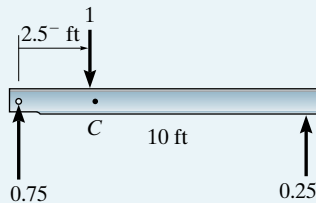
Construct the influence line for the shear at point  $C$  of the beam in Fig. 6–3a.

**SOLUTION**

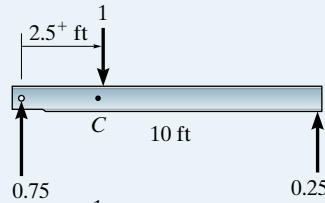
**Tabulate Values.** At each selected position  $x$  of the unit load, the method of sections is used to calculate the value of  $V_C$ . Note in particular that the unit load must be placed *just to the left* ( $x = 2.5^-$ ) and *just to the right* ( $x = 2.5^+$ ) of point  $C$  since the shear is discontinuous at  $C$ , Figs. 6–3b and 6–3c. A plot of the values in Fig. 6–3d yields the influence line for the shear at  $C$ , Fig. 6–3e. Here the diagram gives the shear at  $C$  as the unit load moves from one position to the next along the beam.



(a)

**Fig. 6–3**

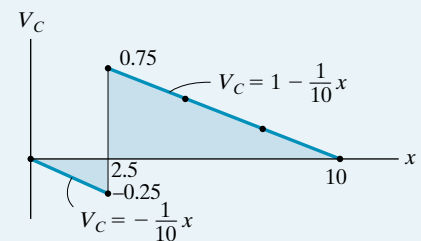
(b)



(c)

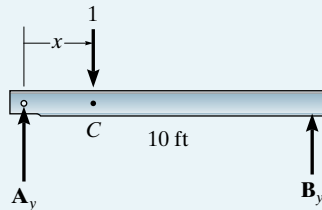
| $x$     | $V_C$ |
|---------|-------|
| 0       | 0     |
| $2.5^-$ | -0.25 |
| $2.5^+$ | 0.75  |
| 5       | 0.5   |
| 7.5     | 0.25  |
| 10      | 0     |

(d)



(e)

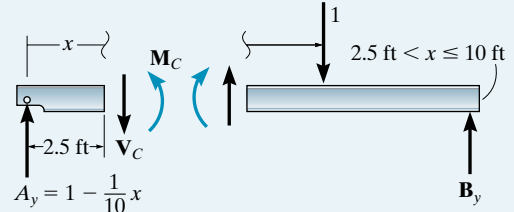
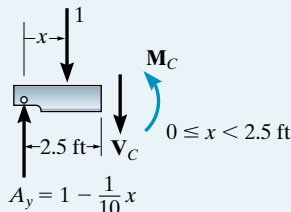
**Influence-Line Equations.** Here two equations have to be determined, since there are two segments for the influence line due to the discontinuity of shear at  $C$ . First we determine  $A_y$ , Fig. 6–3f. Using the result we then obtain the internal shear at  $C$  for each region, Fig. 6–3g. The equations are plotted in Fig. 6–3e.



$$\downarrow + \Sigma M_B = 0; -A_y(10) + 1(10 - x) = 0 \quad \uparrow + \Sigma F_y = 0; 1 - \frac{1}{10}x - 1 - V_C = 0$$

$$A_y = 1 - \frac{1}{10}x \quad V_C = -\frac{1}{10}x$$

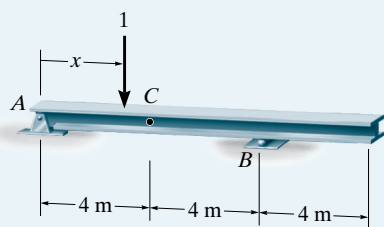
(f)



$$\uparrow + \Sigma F_y = 0; 1 - \frac{1}{10}x - V_C = 0 \quad V_C = 1 - \frac{1}{10}x$$

(g)

## EXAMPLE 6.4



(a)

Fig. 6-4

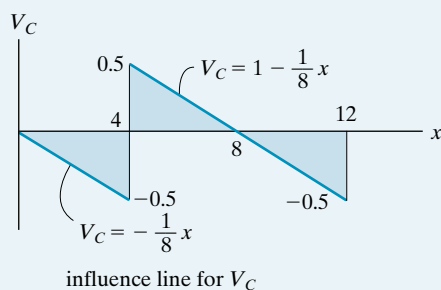
Construct the influence line for the shear at point C of the beam in Fig. 6-4a.

## SOLUTION

**Tabulate Values.** Using statics and the method of sections, verify that the values of the shear  $V_C$  at point C in Fig. 6-4b correspond to each position  $x$  of the unit load on the beam. A plot of the values in Fig. 6-4b yields the influence line in Fig. 6-4c.

| $x$            | $V_C$ |
|----------------|-------|
| 0              | 0     |
| 4 <sup>-</sup> | -0.5  |
| 4 <sup>+</sup> | 0.5   |
| 8              | 0     |
| 12             | -0.5  |

(b)



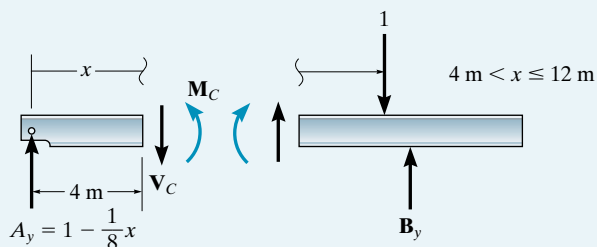
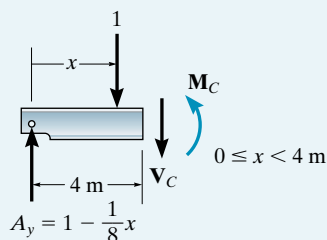
(c)

**Influence-Line Equations.** From Fig. 6-4d, verify that

$$V_C = -\frac{1}{8}x \quad 0 \leq x < 4 \text{ m}$$

$$V_C = 1 - \frac{1}{8}x \quad 4 \text{ m} < x \leq 12 \text{ m}$$

These equations are plotted in Fig. 6-4c.



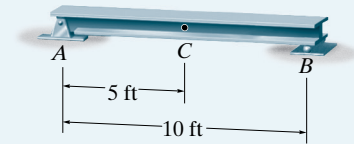
(d)

**EXAMPLE 6.5**

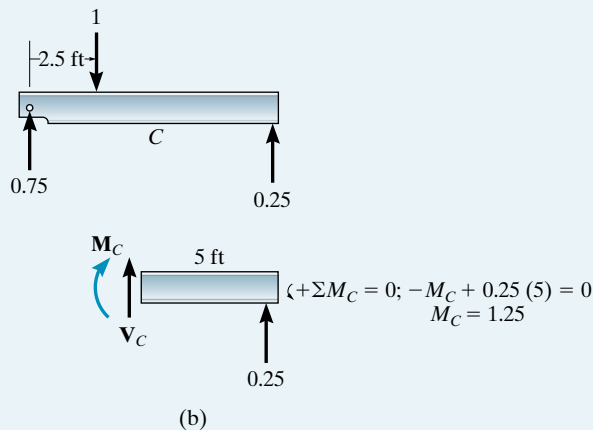
Construct the influence line for the moment at point  $C$  of the beam in Fig. 6–5a.

**SOLUTION**

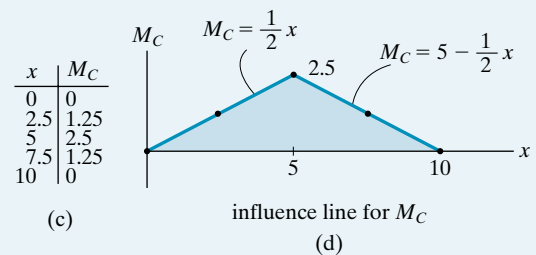
**Tabulate Values.** At each selected position of the unit load, the value of  $M_C$  is calculated using the method of sections. For example, see Fig. 6–5b for  $x = 2.5$  ft. A plot of the values in Fig. 6–5c yields the influence line for the moment at  $C$ , Fig. 6–5d. This diagram gives the moment at  $C$  as the unit load moves from one position to the next along the beam.



(a)

**Fig. 6–5**

(b)



(c)

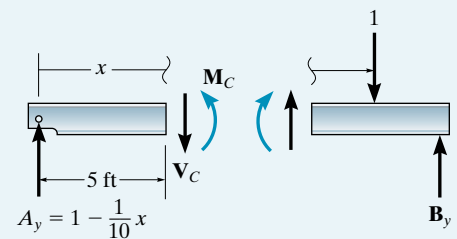
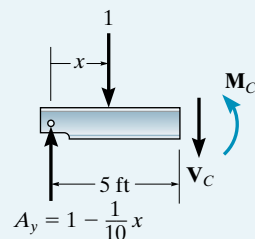
(d)

**Influence-Line Equations.** The two line segments for the influence line can be determined using  $\Sigma M_C = 0$  along with the method of sections shown in Fig. 6–5e. These equations when plotted yield the influence line shown in Fig. 6–5d.

$$\zeta + \Sigma M_C = 0; \quad M_C + 1(5 - x) - \left(1 - \frac{1}{10}x\right)5 = 0 \quad \zeta + \Sigma M_C = 0; \quad M_C - \left(1 - \frac{1}{10}x\right)5 = 0$$

$$M_C = \frac{1}{2}x \quad 0 \leq x < 5 \text{ ft}$$

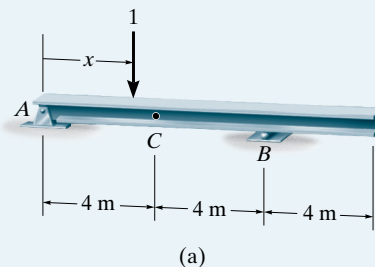
$$M_C = 5 - \frac{1}{2}x \quad 5 \text{ ft} < x \leq 10 \text{ ft}$$



(e)

**EXAMPLE 6.6**

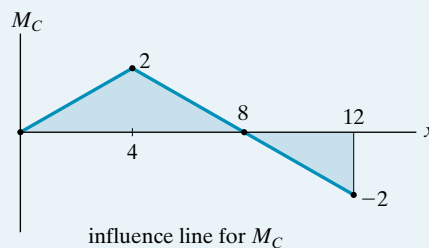
Construct the influence line for the moment at point  $C$  of the beam in Fig. 6-6a.

**Fig. 6-6****SOLUTION**

**Tabulate Values.** Using statics and the method of sections, verify that the values of the moment  $M_C$  at point  $C$  in Fig. 6-6b correspond to each position  $x$  of the unit load. A plot of the values in Fig. 6-6b yields the influence line in Fig. 6-6c.

| $x$ | $M_C$ |
|-----|-------|
| 0   | 0     |
| 4   | 2     |
| 8   | 0     |
| 12  | -2    |

(b)



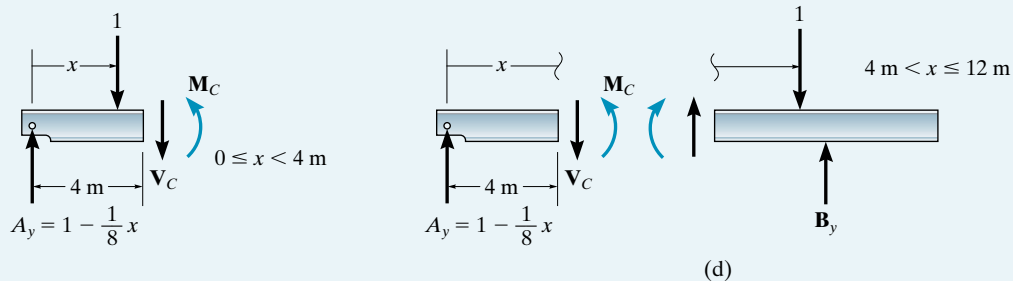
(c)

**Influence-Line Equations.** From Fig. 6-6d verify that

$$M_C = \frac{1}{2}x \quad 0 \leq x < 4 \text{ m}$$

$$M_C = 4 - \frac{1}{2}x \quad 4 \text{ m} < x \leq 12 \text{ m}$$

These equations are plotted in Fig. 6-6c.



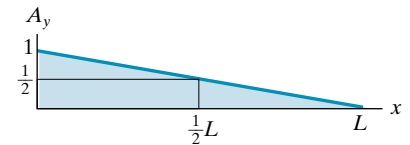
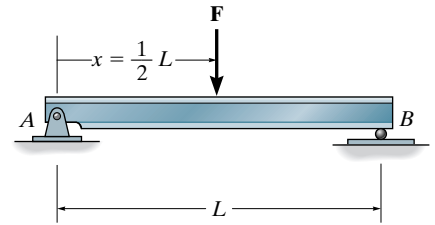
## 6.2 Influence Lines for Beams

Since beams (or girders) often form the main load-carrying elements of a floor system or bridge deck, it is important to be able to construct the influence lines for the reactions, shear, or moment at any specified point in a beam.

**Loadings.** Once the influence line for a function (reaction, shear, or moment) has been constructed, it will then be possible to position the live loads on the beam which will produce the maximum value of the function. Two types of loadings will now be considered.

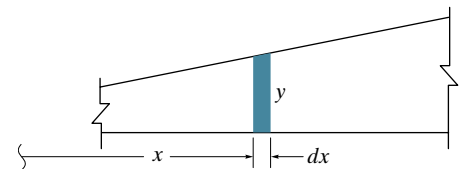
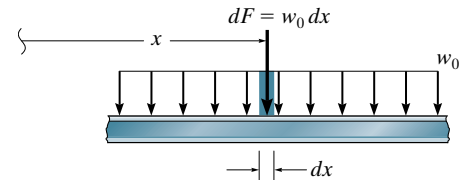
**Concentrated Force.** Since the numerical values of a function for an influence line are determined using a dimensionless unit load, then for any concentrated force  $\mathbf{F}$  acting on the beam at any position  $x$ , the value of the function can be found by multiplying the ordinate of the influence line at the position  $x$  by the magnitude of  $\mathbf{F}$ . For example, consider the influence line for the reaction at  $A$  for the beam  $AB$ , Fig. 6–7. If the unit load is at  $x = \frac{1}{2}L$ , the reaction at  $A$  is  $A_y = \frac{1}{2}$  as indicated from the influence line. Hence, if the force  $F$  lb is at this same point, the reaction is  $A_y = (\frac{1}{2})(F)$  lb. Of course, this same value can also be determined by statics. Obviously, the maximum influence caused by  $\mathbf{F}$  occurs when it is placed on the beam at the same location as the peak of the influence line—in this case at  $x = 0$ , where the reaction would be  $A_y = (1)(F)$  lb.

**Uniform Load.** Consider a portion of a beam subjected to a uniform load  $w_0$ , Fig. 6–8. As shown, each  $dx$  segment of this load creates a concentrated force of  $dF = w_0 dx$  on the beam. If  $dF$  is located at  $x$ , where the beam's influence-line ordinate for some function (reaction, shear, moment) is  $y$ , then the value of the function is  $(dF)(y) = (w_0 dx)y$ . The effect of all the concentrated forces  $dF$  is determined by integrating over the entire length of the beam, that is,  $\int w_0 y dx = w_0 \int y dx$ . Also, since  $\int y dx$  is equivalent to the area under the influence line, then, in general, the value of a function caused by a uniform distributed load is simply the area under the influence line for the function multiplied by the intensity of the uniform load. For example, in the case of a uniformly loaded beam shown in Fig. 6–9, the reaction  $A_y$  can be determined from the influence line as  $A_y = (\text{area})(w_0) = [\frac{1}{2}(1)(L)]w_0 = \frac{1}{2}w_0L$ . This value can of course also be determined from statics.



influence line for  $A_y$

Fig. 6–7



influence line for function

Fig. 6–8

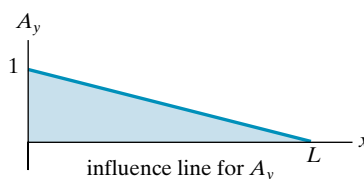
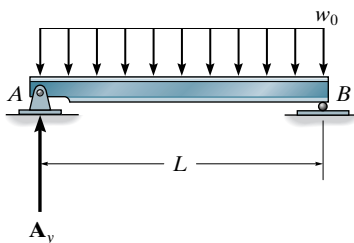


Fig. 6–9

## EXAMPLE 6.7

Determine the maximum *positive* shear that can be developed at point *C* in the beam shown in Fig. 6–10*a* due to a concentrated moving load of 4000 lb and a uniform moving load of 2000 lb/ft.

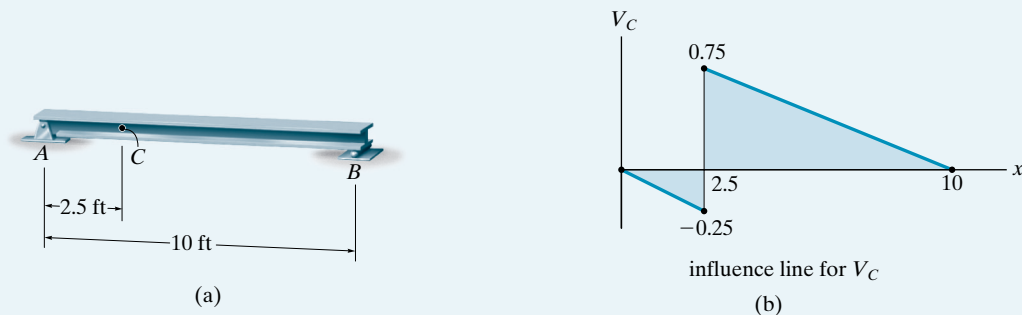


Fig. 6–10

## SOLUTION

The influence line for the shear at *C* has been established in Example 6.3 and is shown in Fig. 6–10*b*.

**Concentrated Force.** The maximum positive shear at *C* will occur when the 4000-lb force is located at  $x = 2.5^+$  ft, since this is the positive peak of the influence line. The ordinate of this peak is +0.75; so that

$$V_C = 0.75(4000 \text{ lb}) = 3000 \text{ lb}$$

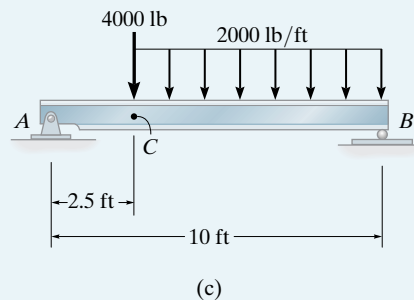
**Uniform Load.** The uniform moving load creates the maximum positive influence for  $V_C$  when the load acts on the beam between  $x = 2.5^+$  ft and  $x = 10$  ft, since within this region the influence line has a positive area. The magnitude of  $V_C$  due to this loading is

$$V_C = \left[ \frac{1}{2}(10 \text{ ft} - 2.5 \text{ ft})(0.75) \right] 2000 \text{ lb/ft} = 5625 \text{ lb}$$

Total Maximum Shear at *C*.

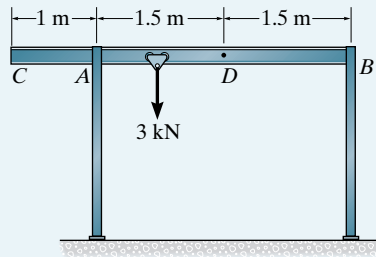
$$(V_C)_{\max} = 3000 \text{ lb} + 5625 \text{ lb} = 8625 \text{ lb} \quad \text{Ans.}$$

Notice that once the *positions* of the loads have been established using the influence line, Fig. 6–10*c*, this value of  $(V_C)_{\max}$  can *also* be determined using statics and the method of sections. Show that this is the case.



**EXAMPLE 6.8**

The frame structure shown in Fig. 6–11a is used to support a hoist for transferring loads for storage at points underneath it. It is anticipated that the load on the dolly is 3 kN and the beam  $CB$  has a mass of 24 kg/m. Assume the dolly has negligible size and can travel the entire length of the beam. Also, assume  $A$  is a pin and  $B$  is a roller. Determine the maximum vertical support reactions at  $A$  and  $B$  and the maximum moment in the beam at  $D$ .



(a)

**SOLUTION**

**Maximum Reaction at A.** We first draw the influence line for  $A_y$ , Fig. 6–11b. Specifically, when a unit load is at  $A$  the reaction at  $A$  is 1 as shown. The ordinate at  $C$ , is 1.33. Here the maximum value for  $A_y$  occurs when the dolly is at  $C$ . Since the dead load (beam weight) must be placed over the entire length of the beam, we have,

$$\begin{aligned}(A_y)_{\max} &= 3000(1.33) + 24(9.81)\left[\frac{1}{2}(4)(1.33)\right] \\ &= 4.63 \text{ kN}\end{aligned}$$

*Ans.*

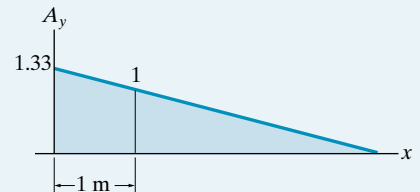
**Maximum Reaction at B.** The influence line (or beam) takes the shape shown in Fig. 6–11c. The values at  $C$  and  $B$  are determined by statics. Here the dolly must be at  $B$ . Thus,

$$\begin{aligned}(B_y)_{\max} &= 3000(1) + 24(9.81)\left[\frac{1}{2}(3)(1)\right] + 24(9.81)\left[\frac{1}{2}(1)(-0.333)\right] \\ &= 3.31 \text{ kN}\end{aligned}$$

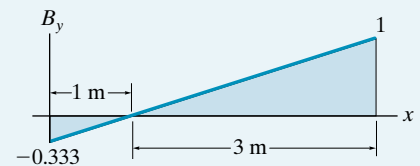
*Ans.*

**Maximum Moment at D.** The influence line has the shape shown in Fig. 6–11d. The values at  $C$  and  $D$  are determined from statics. Here,

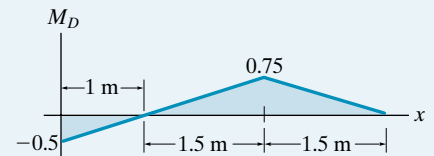
$$\begin{aligned}(M_D)_{\max} &= 3000(0.75) + 24(9.81)\left[\frac{1}{2}(1)(-0.5)\right] + 24(9.81)\left[\frac{1}{2}(3)(0.75)\right] \\ &= 2.46 \text{ kN} \cdot \text{m}\end{aligned}$$

*Ans.*influence line for  $A_y$ 

(b)

influence line for  $B_y$ 

(c)

influence line for  $M_D$ 

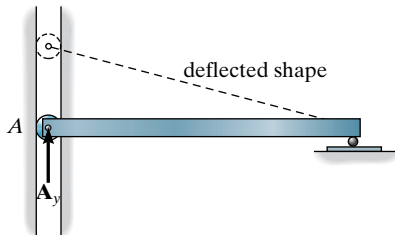
(d)

**Fig. 6–11**

## 6.3 Qualitative Influence Lines



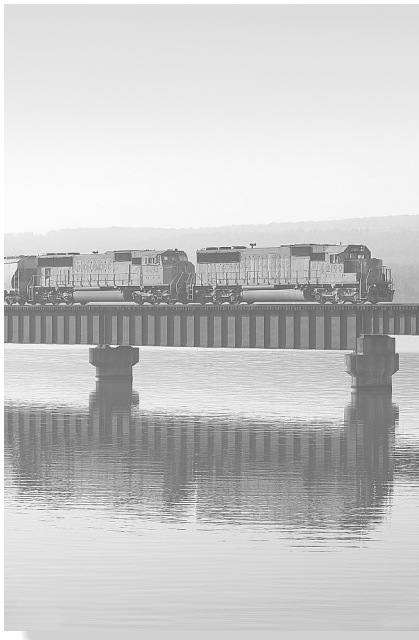
(a)



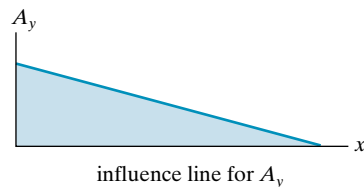
(b)

In 1886, Heinrich Müller-Breslau developed a technique for rapidly constructing the shape of an influence line. Referred to as the **Müller-Breslau principle**, it states that the influence line for a function (reaction, shear, or moment) is to the same scale as the deflected shape of the beam when the beam is acted upon by the function. In order to draw the deflected shape properly, the capacity of the beam to resist the applied function must be removed so the beam can deflect when the function is applied. For example, consider the beam in Fig. 6–12a. If the shape of the influence line for the vertical reaction at  $A$  is to be determined, the pin is first replaced by a roller guide as shown in Fig. 6–12b. A roller guide is necessary since the beam must still resist a horizontal force at  $A$  but no vertical force. When the positive (upward) force  $A_y$  is then applied at  $A$ , the beam deflects to the dashed position,\* which represents the general shape of the influence line for  $A_y$ , Fig. 6–12c. (Numerical values for this specific case have been calculated in Example 6.1.) If the shape of the influence line for the shear at  $C$  is to be determined, Fig. 6–13a, the connection at  $C$  may be symbolized by a roller guide as shown in Fig. 6–13b. This device will resist a moment and axial force but no shear.† Applying a positive shear force  $V_C$  to the beam at  $C$  and allowing the beam to deflect to the dashed position, we find the influence-line shape as shown in Fig. 6–13c. Finally, if the shape of the influence line for the moment at  $C$ , Fig. 6–14a, is to be determined, an internal hinge or pin is placed at  $C$ , since this connection resists axial and shear forces but cannot resist a moment, Fig. 6–14b. Applying positive moments  $M_C$  to the beam, the beam then deflects to the dashed position, which is the shape of the influence line, Fig. 6–14c.

The proof of the Müller-Breslau principle can be established using the principle of virtual work. Recall that *work* is the product of either a linear



Design of this bridge girder is based on influence lines that must be constructed for this train loading.

influence line for  $A_y$ 

(c)

Fig. 6–12

\*Throughout the discussion all deflected positions are drawn to an exaggerated scale to illustrate the concept.

†Here the rollers symbolize supports that carry loads both in tension or compression. See Table 2.1, support (2).

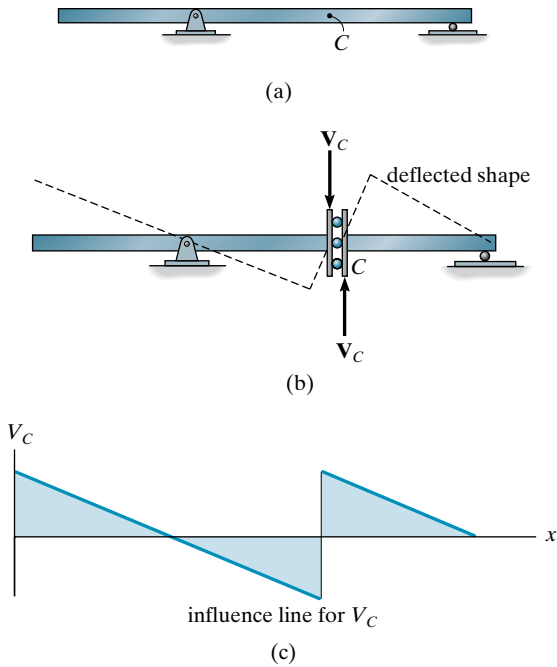


Fig. 6-13

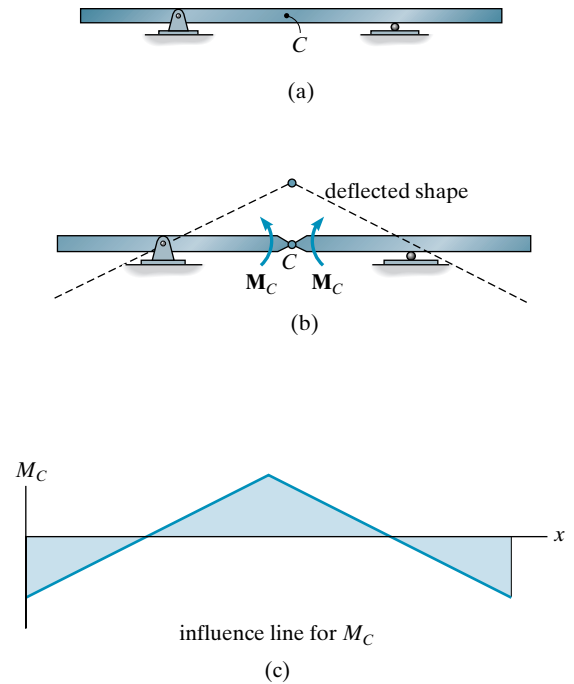


Fig. 6-14

displacement and force in the direction of the displacement or a rotational displacement and moment in the direction of the displacement. If a rigid body (beam) is in equilibrium, the sum of all the forces and moments on it must be equal to zero. Consequently, if the body is given an *imaginary* or *virtual displacement*, the work done by *all* these forces and couple moments must also be equal to zero. Consider, for example, the simply supported beam shown in Fig. 6-15a, which is subjected to a unit load placed at an arbitrary point along its length. If the beam is given a virtual (or imaginary) displacement  $\delta y$  at the support A, Fig. 6-15b, then only the support reaction  $A_y$  and the unit load do virtual work. Specifically,  $A_y$  does positive work  $A_y \delta y$  and the unit load does negative work,  $-1\delta y'$ . (The support at B does not move and therefore the force at B does no work.) Since the beam is in equilibrium and therefore does not actually move, the virtual work sums to zero, i.e.,

$$A_y \delta y - 1 \delta y' = 0$$

If  $\delta y$  is set equal to 1, then

$$A_y = \delta y'$$

In other words, the value of  $A_y$  represents the ordinate of the influence line at the position of the unit load. Since this value is equivalent to the displacement  $\delta y'$  at the position of the unit load, it shows that the *shape* of the influence line for the reaction at A has been established. This proves the Müller-Breslau principle for reactions.

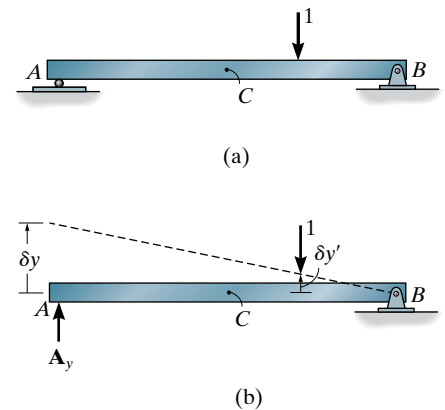


Fig. 6-15

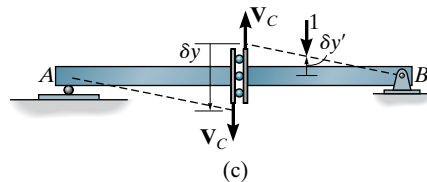


Fig. 6-15

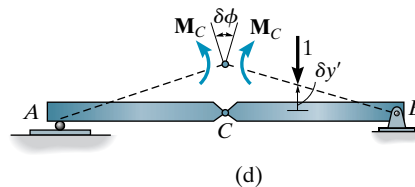
In the same manner, if the beam is sectioned at  $C$ , and the beam undergoes a virtual displacement  $\delta y$  at this point, Fig. 6-15c, then only the internal shear at  $C$  and the unit load do work. Thus, the virtual work equation is

$$V_C \delta y - 1 \delta y' = 0$$

Again, if  $\delta y = 1$ , then

$$V_C = \delta y'$$

and the *shape* of the influence line for the shear at  $C$  has been established.



Lastly, assume a hinge or pin is introduced into the beam at point  $C$ , Fig. 6-15d. If a virtual rotation  $\delta\phi$  is introduced at the pin, virtual work will be done only by the internal moment and the unit load. So

$$M_C \delta\phi - 1 \delta y' = 0$$

Setting  $\delta\phi = 1$ , it is seen that

$$M_C = \delta y'$$

which indicates that the deflected beam has the same *shape* as the influence line for the internal moment at point  $C$  (see Fig. 6-14).

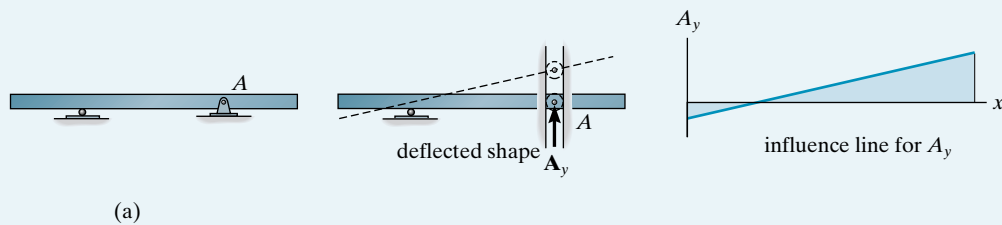
Obviously, the Müller-Breslau principle provides a quick method for establishing the *shape* of the influence line. Once this is known, the ordinates at the peaks can be determined by using the basic method discussed in Sec. 6.1. Also, by simply knowing the general shape of the influence line, it is possible to *locate* the live load on the beam and then determine the maximum value of the function by *using statics*. Example 6.12 illustrates this technique.

**EXAMPLE 6.9**

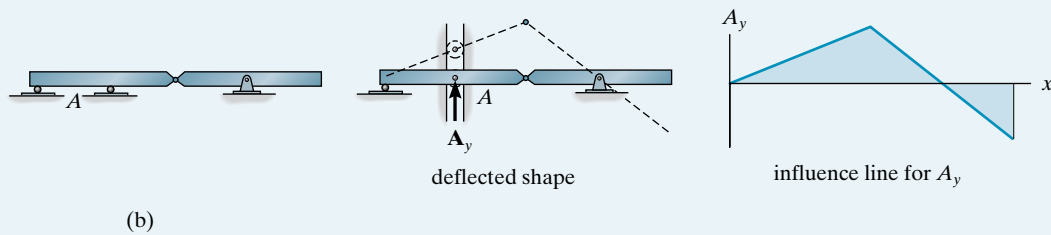
For each beam in Fig. 6–16a through 6–16c, sketch the influence line for the vertical reaction at  $A$ .

**SOLUTION**

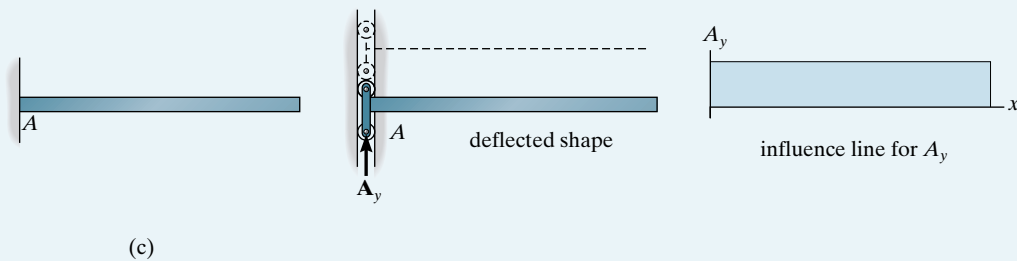
The support is replaced by a roller guide at  $A$  since it will resist  $A_x$ , but not  $A_y$ . The force  $A_y$  is then applied.

**Fig. 6–16**

Again, a roller guide is placed at  $A$  and the force  $A_y$  is applied.



A *double-roller guide* must be used at  $A$  in this case, since this type of support will resist both a moment  $M_A$  at the fixed support and axial load  $A_x$ , but will not resist  $A_y$ .



## EXAMPLE 6.10

For each beam in Figs. 6–17*a* through 6–17*c*, sketch the influence line for the shear at  $B$ .

## SOLUTION

The roller guide is introduced at  $B$  and the positive shear  $V_B$  is applied. Notice that the right segment of the beam will *not* deflect since the roller at  $A$  actually constrains the beam from moving vertically, either up or down. [See support (2) in Table 2.1.]

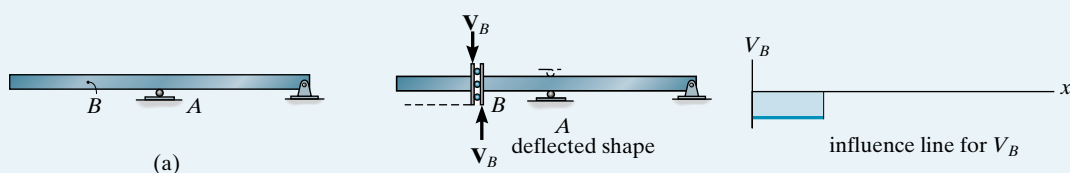
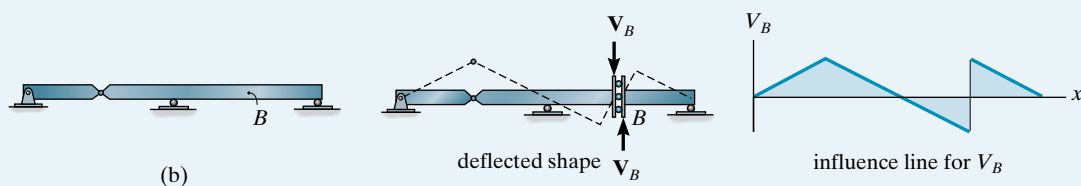
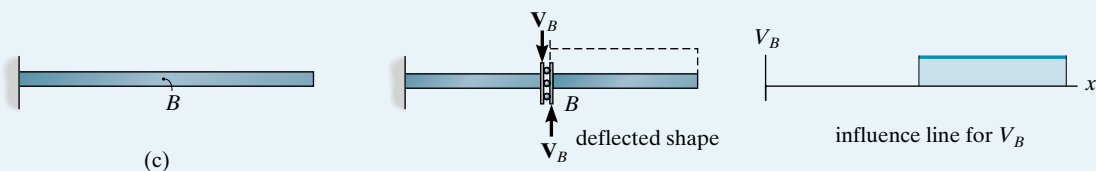


Fig. 6–17

Placing the roller guide at  $B$  and applying the positive shear at  $B$  yields the deflected shape and corresponding influence line.



Again, the roller guide is placed at  $B$ , the positive shear is applied, and the deflected shape and corresponding influence line are shown. Note that the left segment of the beam does not deflect, due to the fixed support.

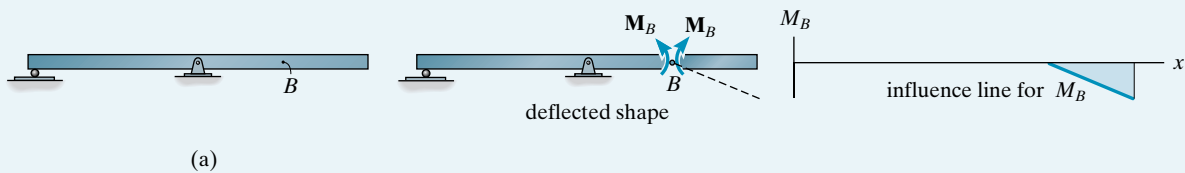


**EXAMPLE 6.11**

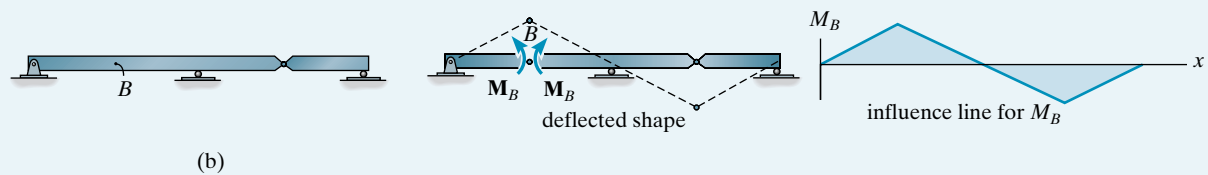
For each beam in Figs. 6–18*a* through Fig. 6–18*c*, sketch the influence line for the moment at  $B$ .

**SOLUTION**

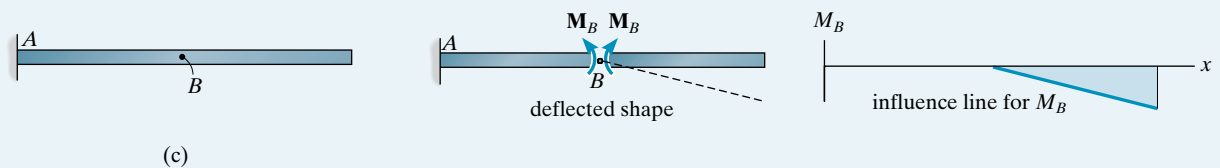
A hinge is introduced at  $B$  and positive moments  $M_B$  are applied to the beam. The deflected shape and corresponding influence line are shown.

**Fig. 6–18**

Placing a hinge at  $B$  and applying positive moments  $M_B$  to the beam yields the deflected shape and influence line.



With the hinge and positive moment at  $B$  the deflected shape and influence line are shown. The left segment of the beam is constrained from moving due to the fixed wall at  $A$ .



## EXAMPLE 6.12

Determine the maximum positive moment that can be developed at point  $D$  in the beam shown in Fig. 6–19a due to a concentrated moving load of 4000 lb, a uniform moving load of 300 lb/ft, and a beam weight of 200 lb/ft.

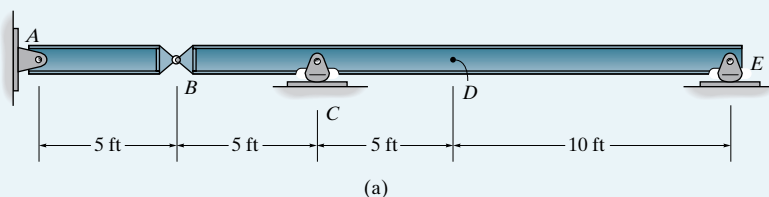


Fig. 6–19

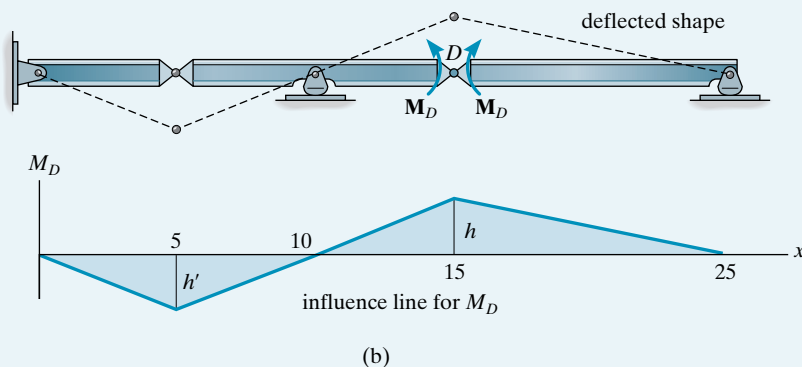
## SOLUTION

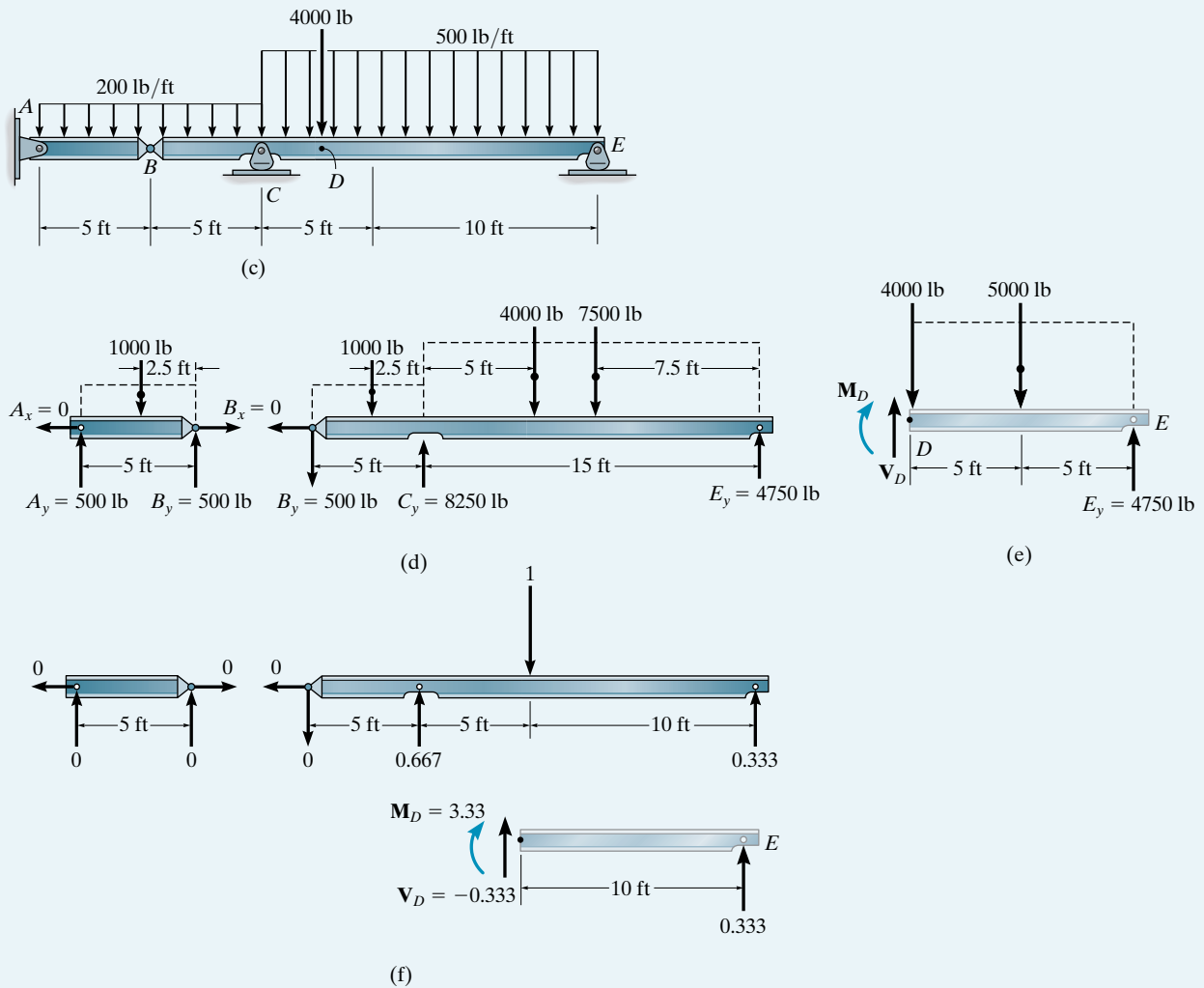
A hinge is placed at  $D$  and positive moments  $M_D$  are applied to the beam. The deflected shape and corresponding influence line are shown in Fig. 6–19b. Immediately one recognizes that the concentrated moving load of 4000 lb creates a maximum *positive* moment at  $D$  when it is placed at  $D$ , i.e., the peak of the influence line. Also, the uniform moving load of 300 lb/ft must extend from  $C$  to  $E$  in order to cover the region where the area of the influence line is positive. Finally, the uniform *dead weight* of 200 lb/ft acts over the *entire length* of the beam. The loading is shown on the beam in Fig. 6–19c. Knowing the position of the loads, we can now determine the maximum moment at  $D$  using statics. In Fig. 6–19d the reactions on  $BE$  have been calculated. Sectioning the beam at  $D$  and using segment  $DE$ , Fig. 6–19e, we have

$$\zeta + \sum M_D = 0; \quad -M_D - 5000(5) + 4750(10) = 0$$

$$M_D = 22\,500 \text{ lb} \cdot \text{ft} = 22.5 \text{ k} \cdot \text{ft}$$

Ans.





This problem can also be worked by using *numerical values* for the influence line as in Sec. 6.1. Actually, by inspection of Fig. 6-19b, only the peak value  $h$  at  $D$  must be determined. This requires placing a unit load on the beam at  $D$  in Fig. 6-19a and then solving for the internal moment in the beam at  $D$ . The calculations are shown in Fig. 6-19f. Thus  $M_D = h = 3.33$ . By proportional triangles,  $h'/(10 - 5) = 3.33/(15 - 10)$  or  $h' = 3.33$ . Hence, with the loading on the beam as in Fig. 6-19c, using the areas and peak values of the influence line, Fig. 6-19b, we have

$$\begin{aligned}
 M_D &= 500 \left[ \frac{1}{2} (25 - 10)(3.33) \right] + 4000(3.33) - 200 \left[ \frac{1}{2} (10)(3.33) \right] \\
 &= 22\,500 \text{ lb} \cdot \text{ft} = 22.5 \text{ k} \cdot \text{ft}
 \end{aligned}$$

*Ans.*

## 6.4 Influence Lines for Floor Girders

Occasionally, floor systems are constructed as shown in Fig. 6–20a, where it can be seen that floor loads are transmitted from **slabs** to **floor beams**, then to **side girders**, and finally supporting **columns**. An idealized model of this system is shown in plane view, Fig. 6–20b. Here the slab is assumed to be a one-way slab and is segmented into simply supported spans resting on the floor beams. Furthermore, the girder is simply supported on the columns. Since the girders are main load-carrying members in this system, it is sometimes necessary to construct their shear and moment influence lines. This is especially true for industrial buildings subjected to heavy concentrated loads. In this regard, notice that a unit load on the floor slab is transferred to the girder only at points where it is in contact with the floor beams, i.e., points *A*, *B*, *C*, and *D*. These points are called **panel points**, and the region between these points is called a **panel**, such as *BC* in Fig. 6–20b.

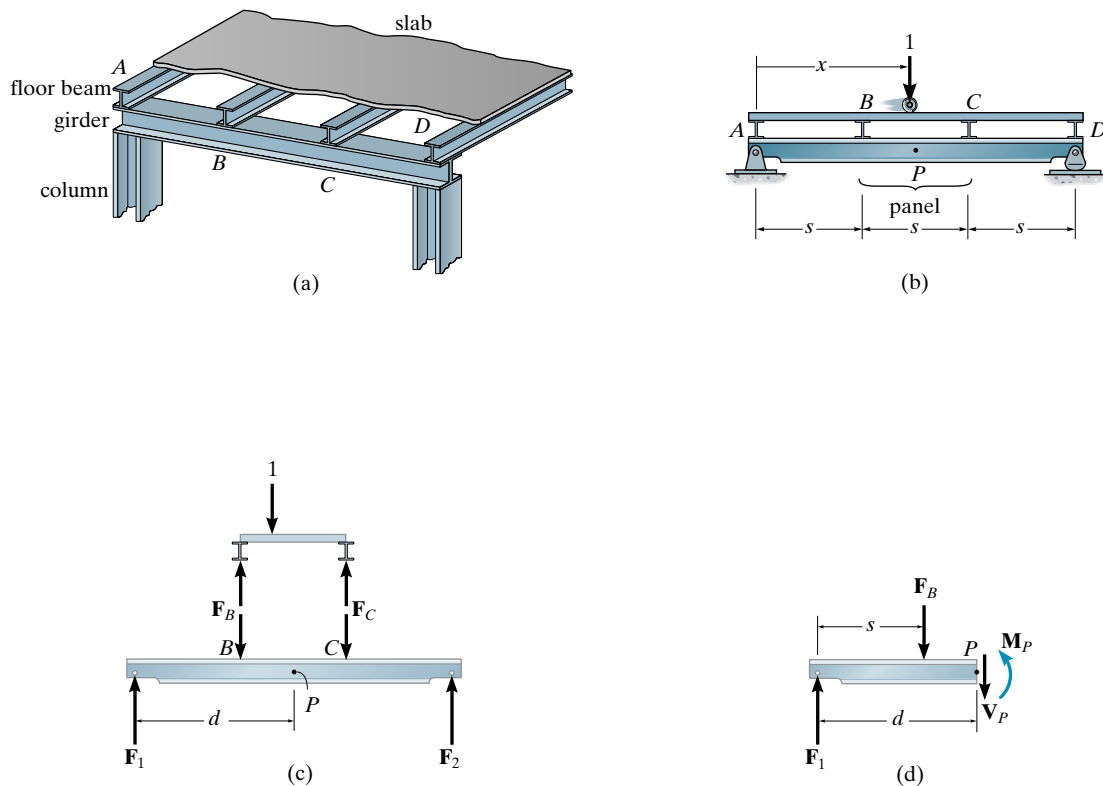


Fig. 6–20

The influence line for a specified point on the girder can be determined using the same statics procedure as in Sec. 6.1; i.e., place the unit load at various points  $x$  on the floor slab and always compute the function (shear or moment) at the specified point  $P$  in the girder, Fig. 6–20*b*. Plotting these values versus  $x$  yields the influence line for the function at  $P$ . In particular, the value for the internal moment in a girder panel will depend upon where point  $P$  is chosen for the influence line, since the magnitude of  $M_P$  depends upon the point's location from the end of the girder. For example, if the unit load acts on the floor slab as shown in Fig. 6–20*c*, one first finds the reactions  $F_B$  and  $F_C$  on the slab, then calculates the support reactions  $F_1$  and  $F_2$  on the girder. The internal moment at  $P$  is then determined by the method of sections, Fig. 6–20*d*. This gives  $M_P = F_1 d - F_B(d - s)$ . Using a similar analysis, the internal shear  $V_P$  can be determined. In this case, however,  $V_P$  will be *constant* throughout the panel  $BC$  ( $V_P = F_1 - F_B$ ) and so it does not depend upon the exact location  $d$  of  $P$  within the panel. For this reason, influence lines for shear in floor girders are specified for *panels* in the girder and not specific points along the girder. The shear is then referred to as *panel shear*. It should also be noted that since the girder is affected only by the loadings transmitted by the floor beams, the unit load is generally placed at each floor-beam location to establish the necessary data used to draw the influence line.

The following numerical examples should clarify the force analysis.



© Juice Images/Alamy

The design of the floor system of this warehouse building must account for critical locations of storage materials on the floor. Influence lines must be used for this purpose. (Photo courtesy of Portland Cement Association.)

## EXAMPLE 6.13

Draw the influence line for the shear in panel  $CD$  of the floor girder in Fig. 6-21a.

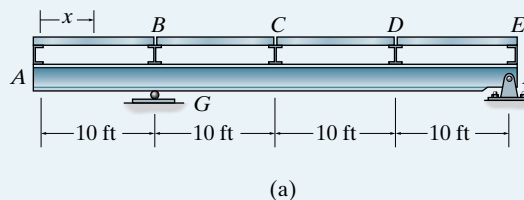


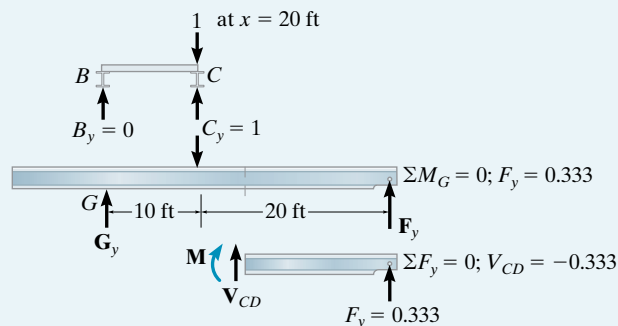
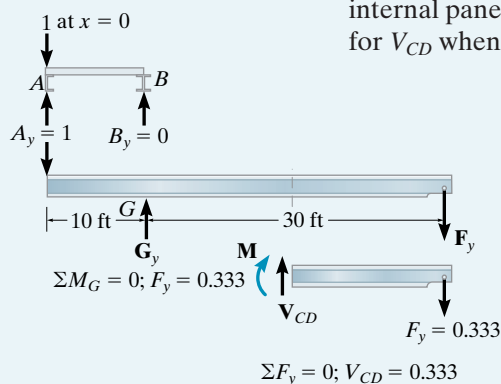
Fig. 6-21

## SOLUTION

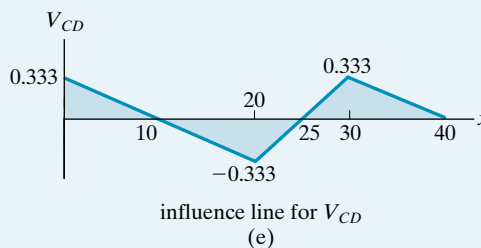
**Tabulate Values.** The unit load is placed at each floor beam location and the shear in panel  $CD$  is calculated. A table of the results is shown in Fig. 6-21b. The details for the calculations when  $x = 0$  and  $x = 20$  ft are given in Figs. 6-21c and 6-21d, respectively. Notice how in each case the reactions of the floor beams on the girder are calculated first, followed by a determination of the girder support reaction at  $F$  ( $G_y$  is not needed), and finally, a segment of the girder is considered and the internal panel shear  $V_{CD}$  is calculated. As an exercise, verify the values for  $V_{CD}$  when  $x = 10$  ft, 30 ft, and 40 ft.

| $x$ | $V_{CD}$ |
|-----|----------|
| 0   | 0.333    |
| 10  | 0        |
| 20  | -0.333   |
| 30  | 0.333    |
| 40  | 0        |

(b)

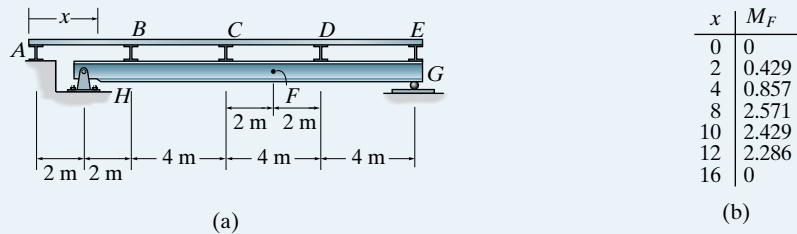


**Influence Line.** When the tabular values are plotted and the points connected with straight line segments, the resulting influence line for  $V_{CD}$  is as shown in Fig. 6-21e.



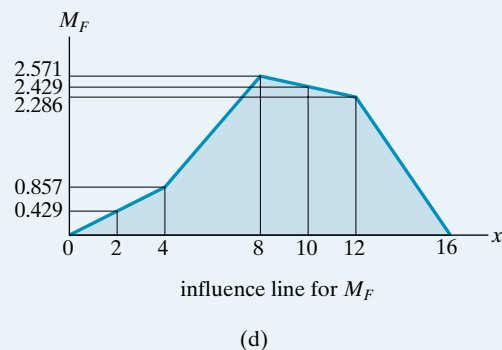
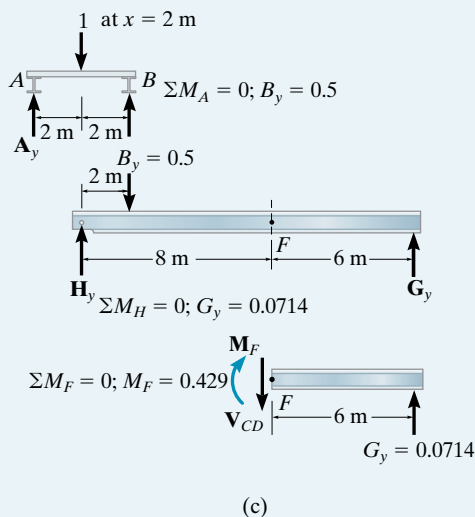
**EXAMPLE 6.14**

Draw the influence line for the moment at point  $F$  for the floor girder in Fig. 6-22a.

**Fig. 6-22****SOLUTION**

**Tabulate Values.** The unit load is placed at  $x = 0$  and each panel point thereafter. The corresponding values for  $M_F$  are calculated and shown in the table, Fig. 6-22b. Details of the calculations for  $x = 2$  m are shown in Fig. 6-22c. As in the previous example, it is first necessary to determine the reactions of the floor beams on the girder, followed by a determination of the girder support reaction  $G_y$  ( $H_y$  is not needed), and finally segment  $GF$  of the girder is considered and the internal moment  $M_F$  is calculated. As an exercise, determine the other values of  $M_F$  listed in Fig. 6-22b.

**Influence Line.** A plot of the tabular values yields the influence line for  $M_F$ , Fig. 6-22d.



## 6.5 Influence Lines for Trusses



The members of this truss bridge were designed using influence lines in accordance with the AASHTO specifications.

Trusses are often used as primary load-carrying elements for bridges. Hence, for design it is important to be able to construct the influence lines for each of its members. As shown in Fig. 6–23, the loading on the bridge deck is transmitted to stringers, which in turn transmit the loading to floor beams and then to the *joints* along the bottom cord of the truss. Since the truss members are affected only by the joint loading, we can therefore obtain the ordinate values of the influence line for a member by loading each joint along the deck with a unit load and then use the method of joints or the method of sections to calculate the force in the member. The data can be arranged in tabular form, listing “unit load at joint” versus “force in member.” As a convention, if the member force is *tensile* it is considered a *positive* value; if it is *compressive* it is *negative*. The influence line for the member is constructed by plotting the data and drawing straight lines between the points.

The following examples illustrate the method of construction.

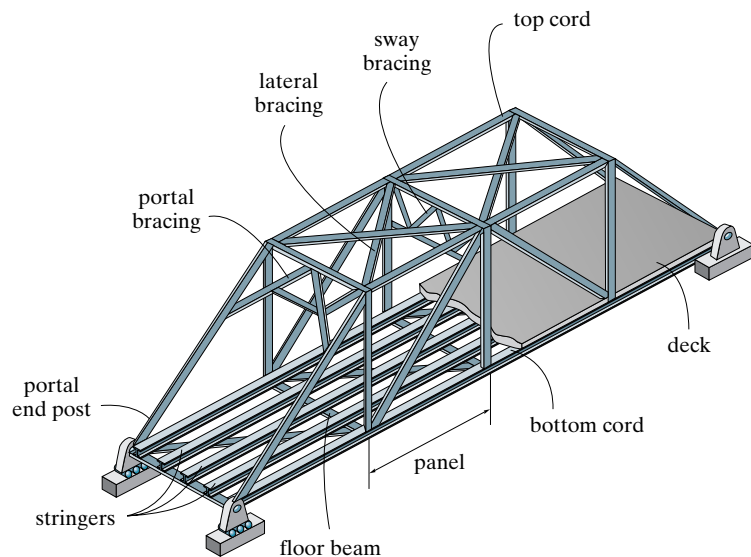
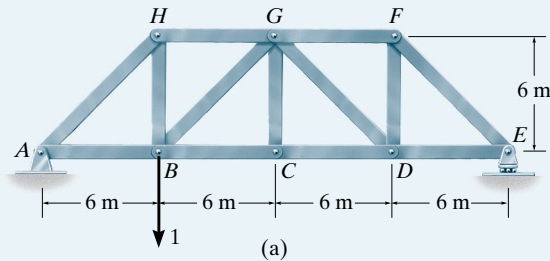


Fig. 6–23

**EXAMPLE 6.15**

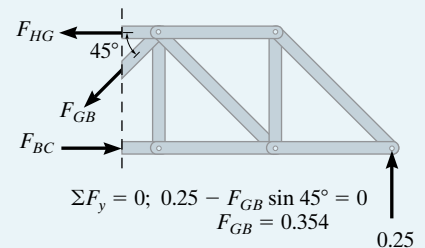
Draw the influence line for the force in member  $GB$  of the bridge truss shown in Fig. 6-24a.

**Fig. 6-24****SOLUTION**

**Tabulate Values.** Here each successive joint at the bottom cord is loaded with a unit load and the force in member  $GB$  is calculated using the method of sections, Fig. 6-24b. For example, placing the unit load at  $x = 6$  m (joint  $B$ ), the support reaction at  $E$  is calculated first, Fig. 6-24a, then passing a section through  $HG$ ,  $GB$ ,  $BC$  and isolating the right segment, the force in  $GB$  is determined, Fig. 6-24c. In the same manner, determine the other values listed in the table.

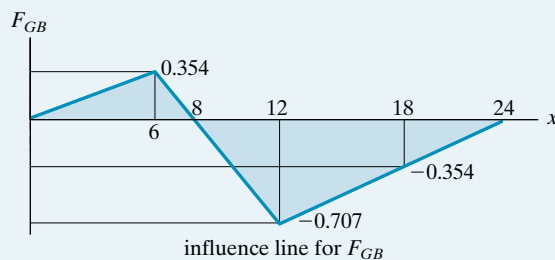
| $x$ | $F_{GB}$ |
|-----|----------|
| 0   | 0        |
| 6   | 0.354    |
| 12  | -0.707   |
| 18  | -0.354   |
| 24  | 0        |

(b)



(c)

**Influence Line.** Plotting the tabular data and connecting the points yields the influence line for member  $GB$ , Fig. 6-24d. Since the influence line extends over the entire span of the truss, member  $GB$  is referred to as a *primary member*. This means  $GB$  is subjected to a force regardless of where the bridge deck (roadway) is loaded, except, of course, at  $x = 8$  m. The point of zero force,  $x = 8$  m, is determined by similar triangles between  $x = 6$  m and  $x = 12$  m, that is,  $(0.354 + 0.707)/(12 - 6) = 0.354/x'$ ,  $x' = 2$  m, so  $x = 6 + 2 = 8$  m.



(d)

## EXAMPLE 6.16

Draw the influence line for the force in member  $CG$  of the bridge truss shown in Fig. 6-25a.

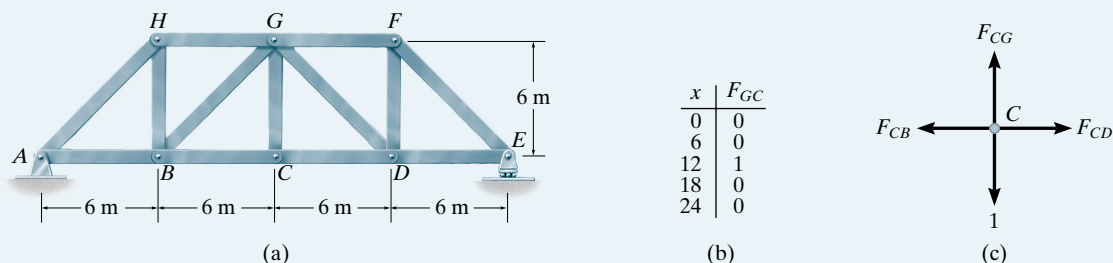
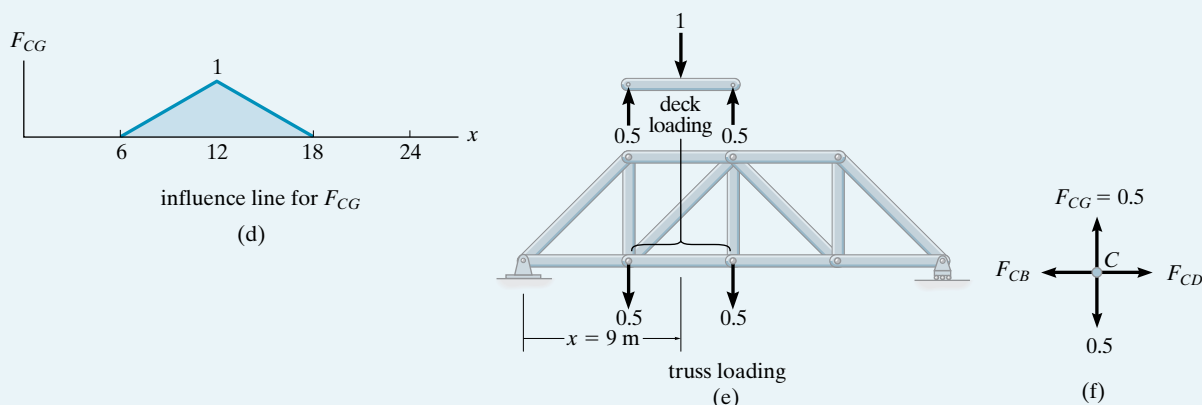


Fig. 6-25

## SOLUTION

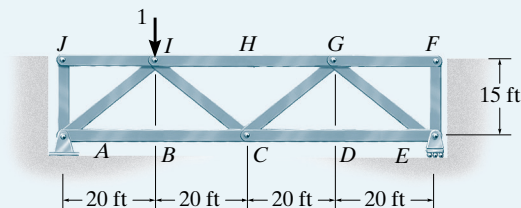
**Tabulate Values.** A table of unit-load position at the joints of the bottom cord versus the force in member  $CG$  is shown in Fig. 6-25b. These values are easily obtained by isolating joint  $C$ , Fig. 6-25c. Here it is seen that  $CG$  is a zero-force member unless the unit load is applied at joint  $C$ , in which case  $F_{CG} = 1$  (T).

**Influence Line.** Plotting the tabular data and connecting the points yields the influence line for member  $CG$  as shown in Fig. 6-25d. In particular, notice that when the unit load is at  $x = 9$  m, the force in member  $CG$  is  $F_{CG} = 0.5$ . This situation requires the unit load to be placed on the bridge deck *between* the joints. The transference of this load from the deck to the truss is shown in Fig. 6-25e. From this one can see that indeed  $F_{CG} = 0.5$  by analyzing the equilibrium of joint  $C$ , Fig. 6-25f. Since the influence line for  $CG$  does *not* extend over the entire span of the truss, Fig. 6-25d, member  $CG$  is referred to as a *secondary member*.



**EXAMPLE 6.17**

In order to determine the maximum force in each member of the Warren truss, shown in the photo, we must first draw the influence lines for each of its members. If we consider a similar truss as shown in Fig. 6–26a, determine the largest force that can be developed in member  $BC$  due to a moving force of 25 k and a moving distributed load of 0.6 k/ft. The loading is applied at the top cord.



(a)

| $x$ | $F_{BC}$ |
|-----|----------|
| 0   | 0        |
| 20  | 1        |
| 40  | 0.667    |
| 60  | 0.333    |
| 80  | 0        |

(b)

**Fig. 6–26****SOLUTION**

**Tabulate Values.** A table of unit-load position  $x$  at the joints along the top cord versus the force in member  $BC$  is shown in Fig. 6–26b. The method of sections can be used for the calculations. For example, when the unit load is at joint  $I$  ( $x = 20$  ft), Fig. 6–26a, the reaction  $E_y$  is determined first ( $E_y = 0.25$ ). Then the truss is sectioned through  $BC$ ,  $IC$ , and  $HI$ , and the right segment is isolated, Fig. 6–26c. One obtains  $F_{BC}$  by summing moments about point  $I$ , to eliminate  $F_{HI}$  and  $F_{IC}$ . In a similar manner determine the other values in Fig. 6–26b.

**Influence Line.** A plot of the tabular values yields the influence line, Fig. 6–26d. By inspection,  $BC$  is a primary member. Why?

**Concentrated Live Force.** The largest force in member  $BC$  occurs when the moving force of 25 k is placed at  $x = 20$  ft. Thus,

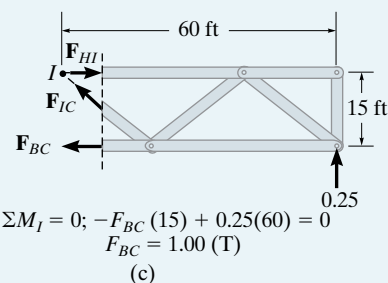
$$F_{BC} = (1.00)(25) = 25.0 \text{ k}$$

**Distributed Live Load.** The uniform live load must be placed over the entire deck of the truss to create the largest tensile force in  $BC$ .\* Thus,

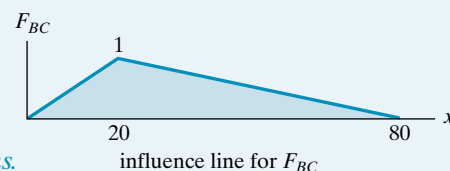
$$F_{BC} = \left[ \frac{1}{2} (80)(1.00) \right] 0.6 = 24.0 \text{ k}$$

Total Maximum Force.

$$(F_{BC})_{\max} = 25.0 \text{ k} + 24.0 \text{ k} = 49.0 \text{ k}$$

**Ans.**

(c)



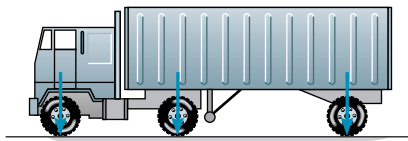
(d)

\*The largest tensile force in member  $GB$  of Example 6.15 is created when the distributed load acts on the deck of the truss from  $x = 0$  to  $x = 8$  m, Fig. 6–24d.

## 6.6 Maximum Influence at a Point due to a Series of Concentrated Loads



As the train passes over this girder bridge the engine and its cars will exert vertical reactions on the girder. These along with the dead load of the bridge must be considered for design.



Once the influence line for a reaction, shear, or moment has been established for a point in a structure, the maximum effect caused by a moving concentrated force is determined by multiplying the peak ordinate of the influence line by the magnitude of the force. In some cases, however, *several* concentrated forces may move along the structure. An example would be the wheel loadings of a truck or train such as shown in Fig. 6–27. In order to determine the maximum effect in this case, either a trial-and-error procedure can be used or a method that is based on the change in the reaction, shear, or moment that takes place as the load is moved. Each of these methods will now be explained.

**Shear.** Consider the simply supported beam with the associated influence line for the shear at point  $C$  in Fig. 6–28a. The maximum *positive shear* at point  $C$  is to be determined due to the series of concentrated (wheel) loads which move from right to left over the beam. The critical loading will occur when one of the loads is placed *just to the right* of point  $C$ , which is coincident with the positive peak of the influence line. By trial and error each of three possible cases can therefore be investigated, Fig. 6–28b. We have

$$\text{Case 1: } (V_C)_1 = 1(0.75) + 4(0.625) + 4(0.5) = 5.25 \text{ k}$$

$$\text{Case 2: } (V_C)_2 = 1(-0.125) + 4(0.75) + 4(0.625) = 5.375 \text{ k}$$

$$\text{Case 3: } (V_C)_3 = 1(0) + 4(-0.125) + 4(0.75) = 2.5 \text{ k}$$

Case 2, with the 1-k force located 5<sup>+</sup> ft from the left support, yields the largest value for  $V_C$  and therefore represents the critical loading. Actually, investigation of Case 3 is unnecessary, since by inspection such an arrangement of loads would yield a value of  $(V_C)_3$  that would be less than  $(V_C)_2$ .

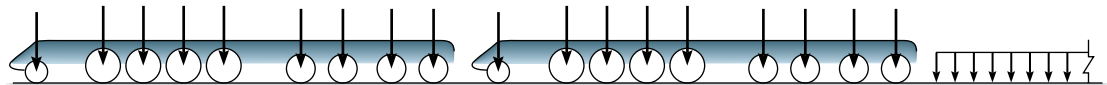
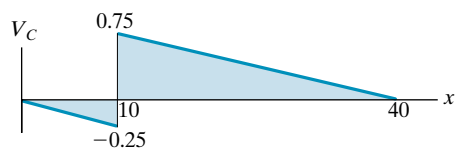
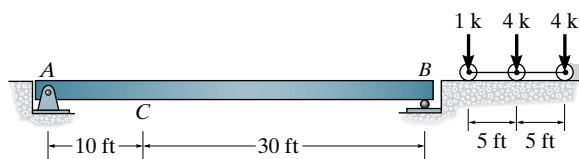
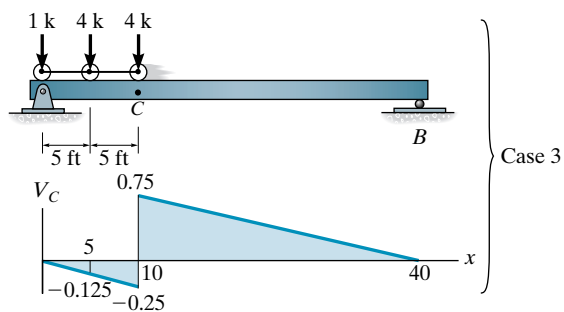
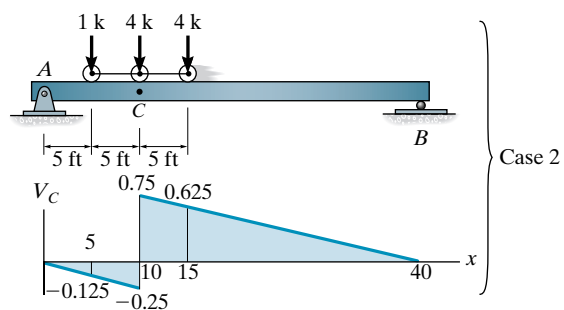
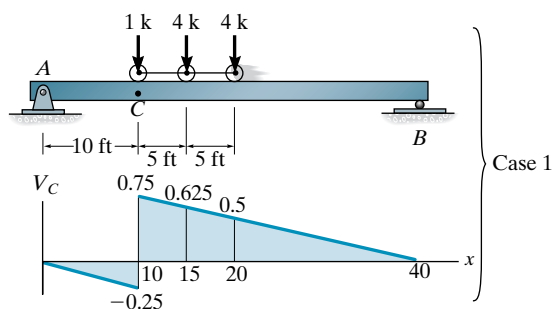


Fig. 6–27



influence line for  $V_C$   
(a)



(b)

Fig. 6-28

When many concentrated loads act on the span, as in the case of the loadings shown in Fig. 6-27, the trial-and-error computations can be tedious. Instead, the critical position of the loads can be determined in a more direct manner by finding the change in shear,  $\Delta V$ , which occurs when the loads move from Case 1 to Case 2, then from Case 2 to Case 3, and so on. As long as each computed  $\Delta V$  is *positive*, the new position will yield a larger shear in the beam at  $C$  than the previous position. Each movement is investigated until a negative change in shear is calculated. When this occurs, the previous position of the loads will give the critical value.

To illustrate this method numerically consider again the beam, loading, and influence line for  $V_C$ , shown in Fig. 6-28a. Since the slope is

$$s = 0.75/(40 - 10) = 0.25/10 = 0.025$$

and the jump at  $C$  has a magnitude of

$$0.75 + 0.25 = 1,$$

then when the loads of Case 1 move 5 ft to Case 2, Fig. 6-28b, the 1-k load jumps *down* ( $-1$ ) and *all* the loads move *up* the slope of the influence line. This causes a change of shear of

$$\Delta V_{1-2} = 1(-1) + [1 + 4 + 4](0.025)(5) = +0.125 \text{ k}$$

Since this result is positive, Case 2 will yield a larger value for  $V_C$  than Case 1. [Compare the answers for  $(V_C)_1$  and  $(V_C)_2$  previously calculated, where indeed  $(V_C)_2 = (V_C)_1 + 0.125$ .] Investigating  $\Delta V_{2-3}$ , which occurs when Case 2 moves to Case 3, Fig. 6-28b, we must account for the downward (negative) jump of the 4-k load and the 5-ft horizontal movement of all the loads *up* the slope of the influence line. We have

$$\Delta V_{2-3} = 4(-1) + (1 + 4 + 4)(0.025)(5) = -2.875 \text{ k}$$

Since  $\Delta V_{2-3}$  is negative, Case 2 is the position of the critical loading, as determined previously.

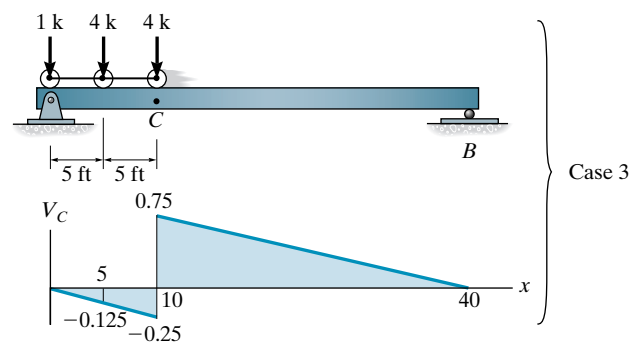
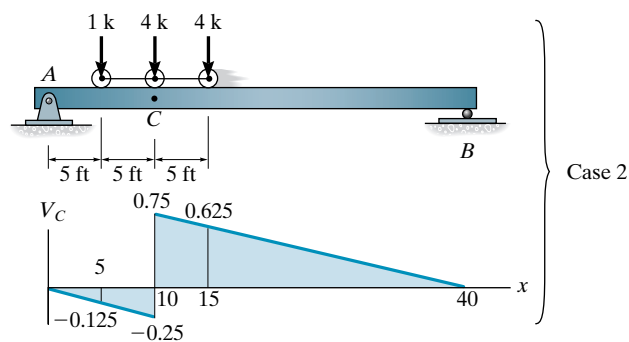
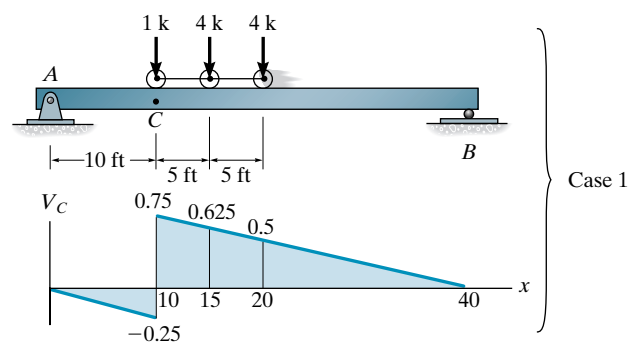
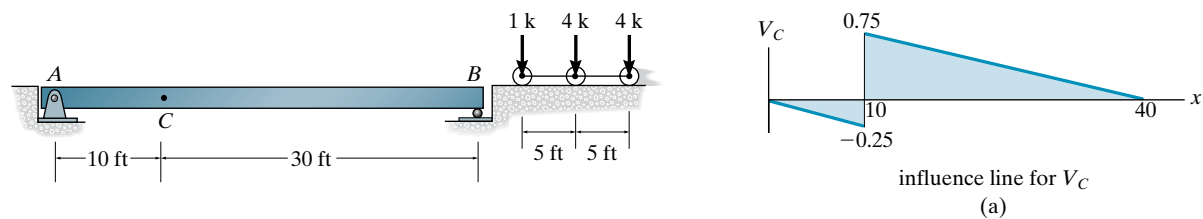


Fig. 6-28 (repeated)

**Moment.** We can also use the foregoing methods to determine the critical position of a series of concentrated forces so that they create the largest internal moment at a specific point in a structure. Of course, it is first necessary to draw the influence line for the moment at the point.

As an example, consider the beam, loading, and influence line for the moment at point  $C$  in Fig. 6–29a. If each of the three concentrated forces is placed on the beam, coincident with the peak of the influence line, we will obtain the greatest influence from each force. The three cases of loading are shown in Fig. 6–29b. When the loads of Case 1 are moved 4 ft to the left to Case 2, it is observed that the 2-k load *decreases*  $\Delta M_{1-2}$ , since the *slope* (7.5/10) is *downward*, Fig. 6–29a. Likewise, the 4-k and 3-k forces cause an *increase* of  $\Delta M_{1-2}$ , since the *slope* [7.5/(40 – 10)] is *upward*. We have

$$\Delta M_{1-2} = -2\left(\frac{7.5}{10}\right)(4) + (4 + 3)\left(\frac{7.5}{40 - 10}\right)(4) = 1.0 \text{ k} \cdot \text{ft}$$

Since  $\Delta M_{1-2}$  is positive, we must further investigate moving the loads 6 ft from Case 2 to Case 3.

$$\Delta M_{2-3} = -(2 + 4)\left(\frac{7.5}{10}\right)(6) + 3\left(\frac{7.5}{40 - 10}\right)(6) = -22.5 \text{ k} \cdot \text{ft}$$

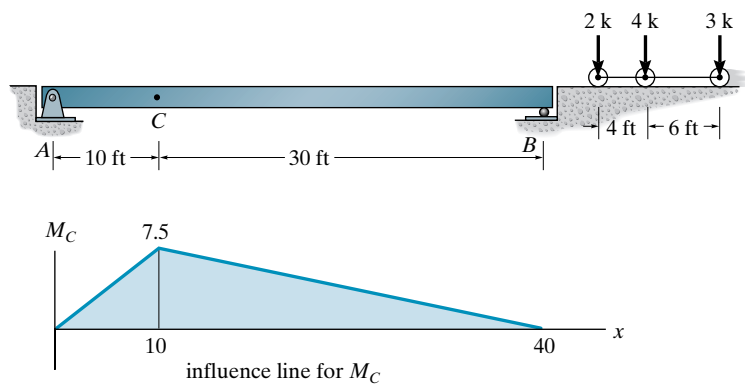
Here the change is negative, so the greatest moment at  $C$  will occur when the beam is loaded as shown in Case 2, Fig. 6–29c. The maximum moment at  $C$  is therefore

$$(M_C)_{\max} = 2(4.5) + 4(7.5) + 3(6.0) = 57.0 \text{ k} \cdot \text{ft}$$

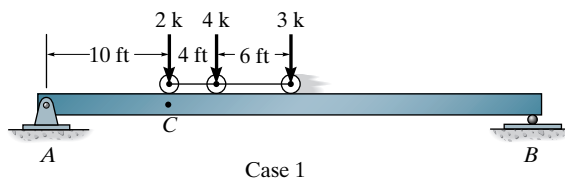
The following examples further illustrate this method.



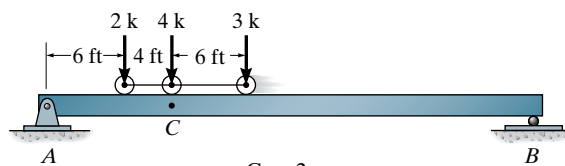
The girders of this bridge must resist the maximum moment caused by the weight of this jet plane as it passes over it.



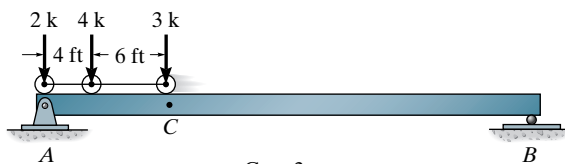
(a)



Case 1

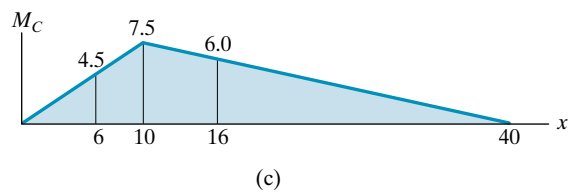


Case 2



Case 3

(b)

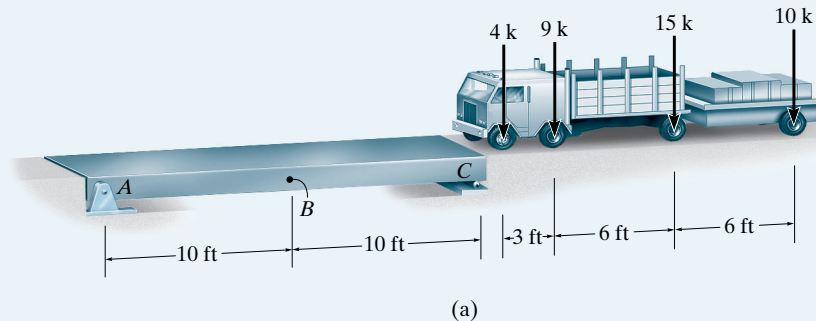


(c)

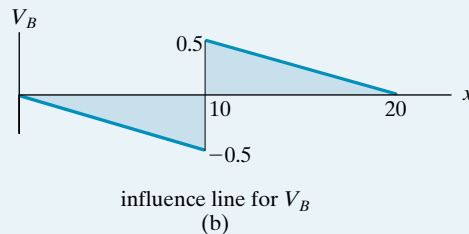
Fig. 6-29

**EXAMPLE 6.18**

Determine the maximum positive shear created at point  $B$  in the beam shown in Fig. 6–30a due to the wheel loads of the moving truck.

**Fig. 6–30****SOLUTION**

The influence line for the shear at  $B$  is shown in Fig. 6–30b.



**3-ft Movement of 4-k Load.** Imagine that the 4-k load acts just to the right of point  $B$  so that we obtain its maximum positive influence. Since the beam segment  $BC$  is 10 ft long, the 10-k load is not as yet on the beam. When the truck moves 3 ft to the left, the 4-k load jumps *downward* on the influence line 1 unit and the 4-k, 9-k, and 15-k loads create a positive increase in  $\Delta V_B$ , since the slope is upward to the left. Although the 10-k load also moves forward 3 ft, it is still not on the beam. Thus,

$$\Delta V_B = 4(-1) + (4 + 9 + 15)\left(\frac{0.5}{10}\right)3 = +0.2 \text{ k}$$

**6-ft Movement of 9-k Load.** When the 9-k load acts just to the right of  $B$ , and then the truck moves 6 ft to the left, we have

$$\Delta V_B = 9(-1) + (4 + 9 + 15)\left(\frac{0.5}{10}\right)(6) + 10\left(\frac{0.5}{10}\right)(4) = +1.4 \text{ k}$$

Note in the calculation that the 10-k load only moves 4 ft on the beam.

**6-ft Movement of 15-k Load.** If the 15-k load is positioned just to the right of  $B$  and then the truck moves 6 ft to the left, the 4-k load moves only 1 ft until it is off the beam, and likewise the 9-k load moves only 4 ft until it is off the beam. Hence,

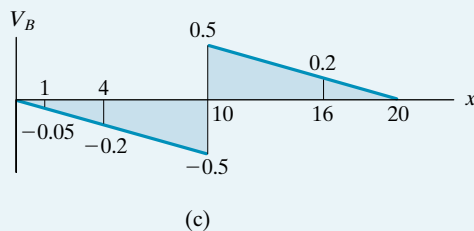
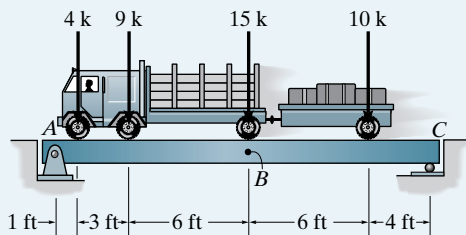
$$\begin{aligned}\Delta V_B &= 15(-1) + 4\left(\frac{0.5}{10}\right)(1) + 9\left(\frac{0.5}{10}\right)(4) + (15 + 10)\left(\frac{0.5}{10}\right)(6) \\ &= -5.5 \text{ k}\end{aligned}$$

Since  $\Delta V_B$  is now negative, the correct position of the loads occurs when the 15-k load is just to the right of point  $B$ , Fig. 6-30c. Consequently,

$$\begin{aligned}(V_B)_{\max} &= 4(-0.05) + 9(-0.2) + 15(0.5) + 10(0.2) \\ &= 7.5 \text{ k}\end{aligned}$$

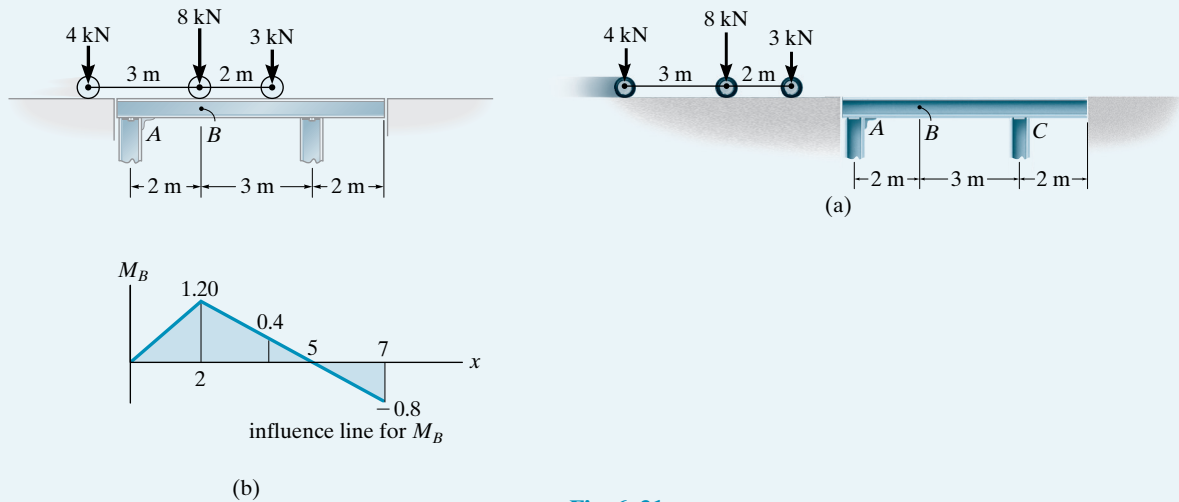
*Ans.*

In practice one also has to consider motion of the truck from left to right and then choose the maximum value between these two situations.



**EXAMPLE 6.19**

Determine the maximum positive moment created at point  $B$  in the beam shown in Fig. 6–31*a* due to the wheel loads of the crane.

**Fig. 6–31****SOLUTION**

The influence line for the moment at  $B$  is shown in Fig. 6–31*b*.

**2-m Movement of 3-kN Load.** If the 3-kN load is assumed to act at  $B$  and then moves 2 m to the right, Fig. 6–31*b*, the change in moment is

$$\Delta M_B = -3\left(\frac{1.20}{3}\right)(2) + 8\left(\frac{1.20}{2}\right)(2) = 7.20 \text{ kN} \cdot \text{m}$$

Why is the 4-kN load not included in the calculations?

**3-m Movement of 8-kN Load.** If the 8-kN load is assumed to act at  $B$  and then moves 3 m to the right, the change in moment is

$$\begin{aligned} \Delta M_B &= -3\left(\frac{1.20}{3}\right)(3) - 8\left(\frac{1.20}{3}\right)(3) + 4\left(\frac{1.20}{2}\right)(2) \\ &= -8.40 \text{ kN} \cdot \text{m} \end{aligned}$$

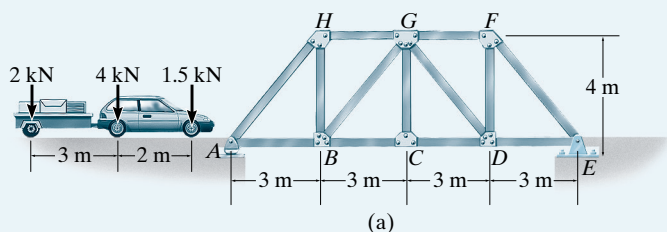
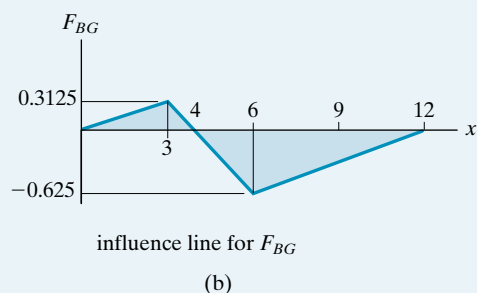
Notice here that the 4-kN load was initially 1 m off the beam, and therefore moves only 2 m on the beam.

Since there is a sign change in  $\Delta M_B$ , the correct position of the loads for maximum positive moment at  $B$  occurs when the 8-kN force is at  $B$ , Fig. 6–31*b*. Therefore,

$$(M_B)_{\max} = 8(1.20) + 3(0.4) = 10.8 \text{ kN} \cdot \text{m} \quad \text{Ans.}$$

**EXAMPLE 6.20**

Determine the maximum compressive force developed in member  $BG$  of the side truss in Fig. 6-32a due to the right side wheel loads of the car and trailer. Assume the loads are applied directly to the truss and move only to the right.

**Fig. 6-32****SOLUTION**

The influence line for the force in member  $BG$  is shown in Fig. 6-32b. Here a trial-and-error approach for the solution will be used. Since we want the greatest negative (compressive) force in  $BG$ , we begin as follows:

**1.5-kN Load at Point C.** In this case

$$\begin{aligned} F_{BG} &= 1.5 \text{ kN}(-0.625) + 4(0) + 2 \text{ kN}\left(\frac{0.3125}{3 \text{ m}}\right)(1 \text{ m}) \\ &= -0.729 \text{ kN} \end{aligned}$$

**4-kN Load at Point C.** By inspection this would seem a more reasonable case than the previous one.

$$\begin{aligned} F_{BG} &= 4 \text{ kN}(-0.625) + 1.5 \text{ kN}\left(\frac{-0.625}{6 \text{ m}}\right)(4 \text{ m}) + 2 \text{ kN}(0.3125) \\ &= -2.50 \text{ kN} \end{aligned}$$

**2-kN Load at Point C.** In this case all loads will create a compressive force in  $BC$ .

$$\begin{aligned} F_{BG} &= 2 \text{ kN}(-0.625) + 4 \text{ kN}\left(\frac{-0.625}{6 \text{ m}}\right)(3 \text{ m}) + 1.5 \text{ kN}\left(\frac{-0.625}{6 \text{ m}}\right)(1 \text{ m}) \\ &= -2.66 \text{ kN} \end{aligned}$$

*Ans.*

Since this final case results in the largest answer, the critical loading occurs when the 2-kN load is at C.

## 6.7 Absolute Maximum Shear and Moment

In Sec. 6.6 we developed the methods for computing the maximum shear and moment at a *specified point* in a beam due to a series of concentrated moving loads. A more general problem involves the determination of both the *location of the point* in the beam and the *position of the loading* on the beam so that one can obtain the *absolute maximum* shear and moment caused by the loads. If the beam is cantilevered or simply supported, this problem can be readily solved.

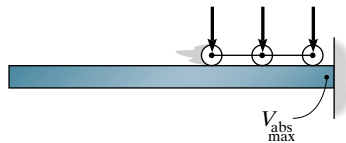


Fig. 6-33

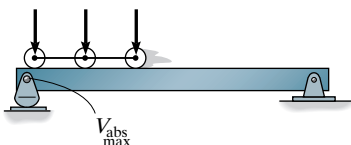


Fig. 6-34

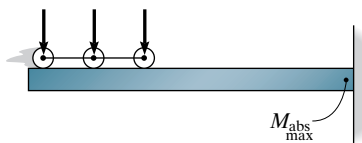


Fig. 6-35

**Shear.** For a *cantilevered beam* the absolute maximum shear will occur at a point located just next to the fixed support. The maximum shear is found by the method of sections, with the loads positioned anywhere on the span, Fig. 6-33.

For *simply supported beams* the absolute maximum shear will occur just next to one of the supports. For example, if the loads are equivalent, they are positioned so that the first one in sequence is placed close to the support, as in Fig. 6-34.

**Moment.** The absolute maximum moment for a *cantilevered beam* occurs at the same point where absolute maximum shear occurs, although in this case the concentrated loads should be positioned at the *far end* of the beam, as in Fig. 6-35.

For a *simply supported beam* the critical position of the loads and the associated absolute maximum moment cannot, in general, be determined by inspection. We can, however, determine the position analytically. For purposes of discussion, consider a beam subjected to the forces  $F_1$ ,  $F_2$ ,  $F_3$  shown in Fig. 6-36a. Since the moment diagram for a series of concentrated forces consists of straight line segments having peaks at each force, the absolute maximum moment will occur under one of the forces. Assume this maximum moment occurs under  $F_2$ . The position of the loads  $F_1$ ,  $F_2$ ,  $F_3$  on the beam will be specified by the distance  $x$ , measured from  $F_2$  to the beam's centerline as shown. To determine a specific value of  $x$ , we first obtain the resultant force of the system,  $F_R$ , and its distance

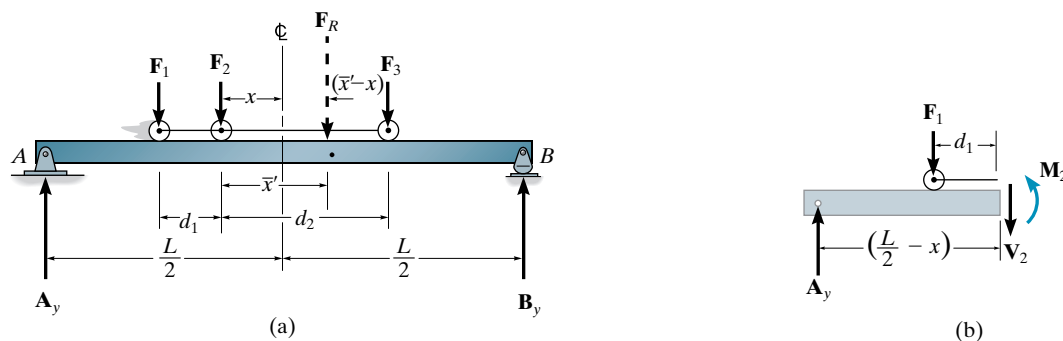


Fig. 6-36

$\bar{x}'$  measured from  $F_2$ . Once this is done, moments are summed about  $B$ , which yields the beam's left reaction,  $A_y$ , that is,

$$\Sigma M_B = 0; \quad A_y = \frac{1}{L} (F_R) \left[ \frac{L}{2} - (\bar{x}' - x) \right]$$

If the beam is sectioned just to the left of  $F_2$ , the resulting free-body diagram is shown in Fig. 6-36b. The moment  $M_2$  under  $F_2$  is therefore

$$\begin{aligned} \Sigma M = 0; \quad M_2 &= A_y \left( \frac{L}{2} - x \right) - F_1 d_1 \\ &= \frac{1}{L} (F_R) \left[ \frac{L}{2} - (\bar{x}' - x) \right] \left( \frac{L}{2} - x \right) - F_1 d_1 \\ &= \frac{F_R L}{4} - \frac{F_R \bar{x}'}{2} - \frac{F_R x^2}{L} + \frac{F_R x \bar{x}'}{L} - F_1 d_1 \end{aligned}$$

For maximum  $M_2$  we require

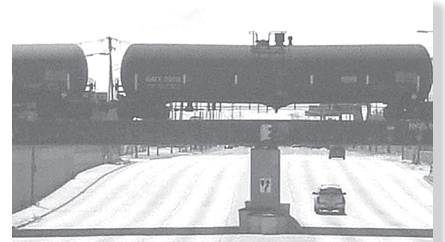
$$\frac{dM_2}{dx} = \frac{-2F_R x}{L} + \frac{F_R \bar{x}'}{L} = 0$$

or

$$x = \frac{\bar{x}'}{2}$$

Hence, we may conclude that the *absolute maximum moment in a simply supported beam occurs under one of the concentrated forces, such that this force is positioned on the beam so that it and the resultant force of the system are equidistant from the beam's centerline*. Since there are a series of loads on the span (for example,  $F_1$ ,  $F_2$ ,  $F_3$  in Fig. 6-36a), this principle will have to be applied to each load in the series and the corresponding maximum moment computed. By comparison, the largest moment is the absolute maximum. As a general rule, though, the absolute maximum moment often occurs under the largest force lying nearest the resultant force of the system.

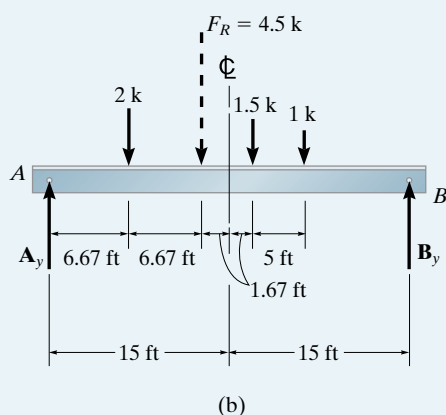
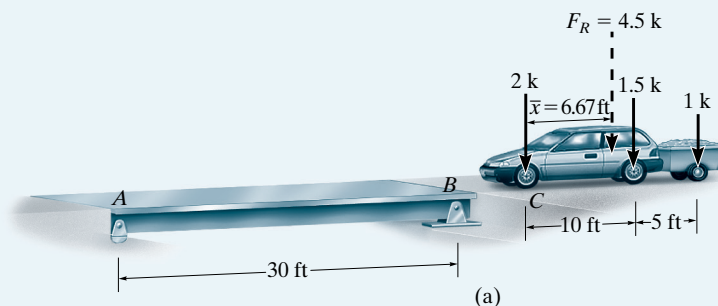
**Envelope of Maximum Influence-Line Values.** Rules or formulations for determining the absolute maximum shear or moment are difficult to establish for beams supported in ways other than the cantilever or simple support discussed here. An elementary way to proceed to solve this problem, however, requires constructing influence lines for the shear or moment at selected points along the entire length of the beam and then computing the maximum shear or moment in the beam for each point using the methods of Sec. 6.6. These values when plotted yield an “envelope of maximums,” from which both the absolute maximum value of shear or moment and its location can be found. Obviously, a computer solution for this problem is desirable for complicated situations, since the work can be rather tedious if carried out by hand calculations.



The absolute maximum moment in this girder bridge is the result of the moving concentrated loads caused by the wheels of these train cars. The cars must be in the critical position, and the location of the point in the girder where the absolute maximum moment occurs must be identified.

**EXAMPLE 6.21**

Determine the absolute maximum moment in the simply supported bridge deck shown in Fig. 6–37a.

**SOLUTION**

The magnitude and position of the resultant force of the system are determined first, Fig. 6–37a. We have

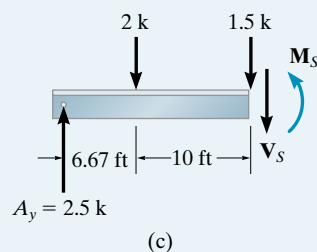
$$\begin{aligned}
 +\downarrow F_R &= \Sigma F; & F_R &= 2 + 1.5 + 1 = 4.5 \text{ k} \\
 \zeta + M_{R_C} &= \Sigma M_C; & 4.5\bar{x} &= 1.5(10) + 1(15) \\
 & & \bar{x} &= 6.67 \text{ ft}
 \end{aligned}$$

Let us first assume the absolute maximum moment occurs under the 1.5-k load. The load and the resultant force are positioned equidistant from the beam's centerline, Fig. 6–37b. Calculating  $A_y$  first, Fig. 6–37b, we have

$$\zeta + \Sigma M_B = 0; \quad -A_y(30) + 4.5(16.67) = 0 \quad A_y = 2.50 \text{ k}$$

Now using the left section of the beam, Fig. 6–37c, yields

$$\begin{aligned}
 \zeta + \Sigma M_S = 0; & \quad -2.50(16.67) + 2(10) + M_S = 0 \\
 & \quad M_S = 21.7 \text{ k} \cdot \text{ft}
 \end{aligned}$$



**Fig. 6–37**

There is a possibility that the absolute maximum moment may occur under the 2-k load, since  $2\text{ k} > 1.5\text{ k}$  and  $F_R$  is between both 2 k and 1.5 k. To investigate this case, the 2-k load and  $F_R$  are positioned equidistant from the beam's centerline, Fig. 6-37d. Show that  $A_y = 1.75\text{ k}$  as indicated in Fig. 6-37e and that

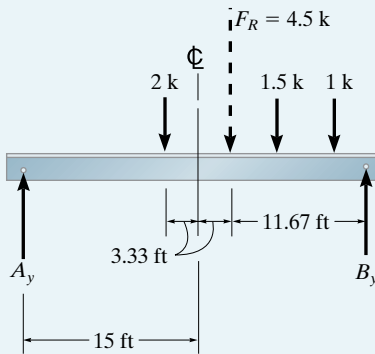
$$M_S = 20.4\text{ k} \cdot \text{ft}$$

By comparison, the absolute maximum moment is

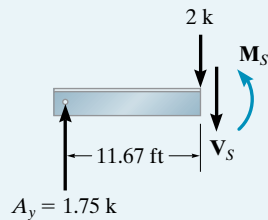
$$M_S = 21.7\text{ k} \cdot \text{ft}$$

*Ans.*

which occurs under the 1.5-k load, when the loads are positioned on the beam as shown in Fig. 6-37b.



(d)



(e)

## EXAMPLE 6.22



The truck has a mass of  $2 \text{ Mg}$  and a center of gravity at  $G$  as shown in Fig. 6–38a. Determine the absolute maximum moment developed in the simply supported bridge deck due to the truck's weight. The bridge has a length of  $10 \text{ m}$ .

## SOLUTION

As noted in Fig. 6–38a, the weight of the truck,  $2(10^3) \text{ kg}(9.81 \text{ m/s}^2) = 19.62 \text{ kN}$ , and the wheel reactions have been calculated by statics. Since the largest reaction occurs at the front wheel, we will select this wheel along with the resultant force and position them *equidistant* from the centerline of the bridge, Fig. 6–38b. Using the resultant force rather than the wheel loads, the vertical reaction at  $B$  is then

$$\begin{aligned} \zeta + \Sigma M_A = 0; \quad & B_y(10) - 19.62(4.5) = 0 \\ & B_y = 8.829 \text{ kN} \end{aligned}$$

The maximum moment occurs under the front wheel loading. Using the right section of the bridge deck, Fig. 6–38c, we have

$$\begin{aligned} \zeta + \Sigma M_S = 0; \quad & 8.829(4.5) - M_s = 0 \\ & M_s = 39.7 \text{ kN} \cdot \text{m} \end{aligned}$$

Ans.

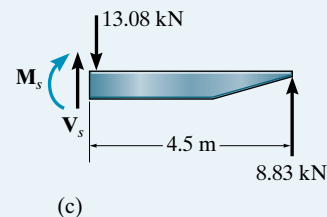
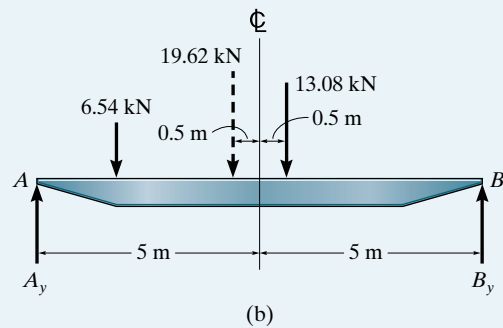
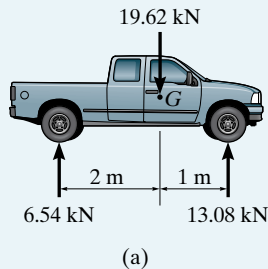


Fig. 6–38

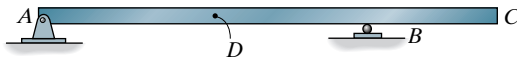
## FUNDAMENTAL PROBLEMS

**F6-1.** Use the Müller-Breslau principle to sketch the influence lines for the vertical reaction at  $A$ , the shear at  $C$ , and the moment at  $C$ .



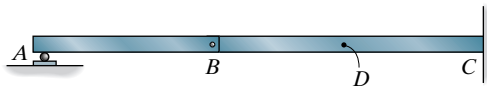
F6-1

**F6-2.** Use the Müller-Breslau principle to sketch the influence lines for the vertical reaction at  $A$ , the shear at  $D$ , and the moment at  $B$ .



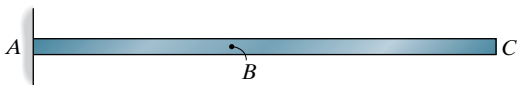
F6-2

**F6-3.** Use the Müller-Breslau principle to sketch the influence lines for the vertical reaction at  $A$ , the shear at  $D$ , and the moment at  $D$ .



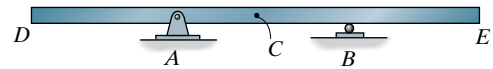
F6-3

**F6-4.** Use the Müller-Breslau principle to sketch the influence lines for the vertical reaction at  $A$ , the shear at  $B$ , and the moment at  $B$ .



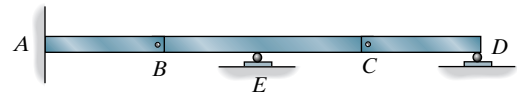
F6-4

**F6-5.** Use the Müller-Breslau principle to sketch the influence lines for the vertical reaction at  $A$ , the shear at  $C$ , and the moment at  $C$ .



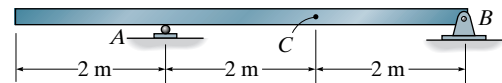
F6-5

**F6-6.** Use the Müller-Breslau principle to sketch the influence lines for the vertical reaction at  $A$ , the shear just to the left of the roller support at  $E$ , and the moment at  $A$ .



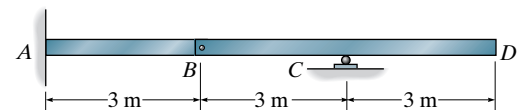
F6-6

**F6-7.** The beam supports a distributed live load of  $1.5 \text{ kN/m}$  and single concentrated load of  $8 \text{ kN}$ . The dead load is  $2 \text{ kN/m}$ . Determine (a) the maximum positive moment at  $C$ , (b) the maximum positive shear at  $C$ .



F6-7

**F6-8.** The beam supports a distributed live load of  $2 \text{ kN/m}$  and single concentrated load of  $6 \text{ kN}$ . The dead load is  $4 \text{ kN/m}$ . Determine (a) the maximum vertical positive reaction at  $C$ , (b) the maximum negative moment at  $A$ .



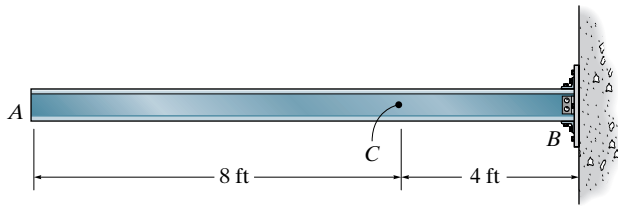
F6-8

## PROBLEMS

### Sec. 6.1–6.3

**6-1.** Draw the influence lines for (a) the moment at  $C$ , (b) the vertical reaction at  $B$ , and (c) the shear at  $C$ . Assume  $B$  is a fixed support. Solve this problem using the basic method of Sec. 6.1.

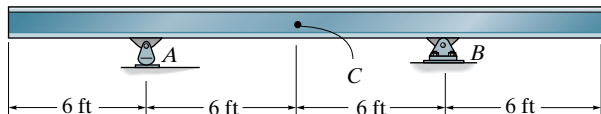
**6-2.** Solve Prob. 6-1 using the Müller-Breslau principle.



**Probs. 6-1/2**

**6** **6-3.** Draw the influence lines for (a) the vertical reaction at  $A$ , (b) the moment at  $C$ , and (c) the shear just to the left of the support at  $B$ . Solve this problem using the basic method of Sec. 6.1.

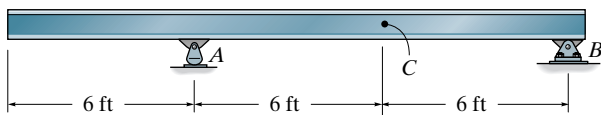
**\*6-4.** Solve Prob. 6-3 using the Müller-Breslau principle.



**Probs. 6-3/4**

**6-5.** Draw the influence lines for (a) the vertical reaction at  $B$ , (b) the shear just to the right of the rocker at  $A$ , and (c) the moment at  $C$ . Solve this problem using the basic method of Sec. 6.1.

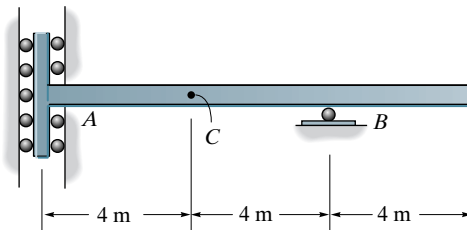
**6-6.** Solve Prob. 6-5 using Müller-Breslau's principle.



**Probs. 6-5/6**

**6-7.** Draw the influence line for (a) the moment at  $B$ , (b) the shear at  $C$ , and (c) the vertical reaction at  $B$ . Solve this problem using the basic method of Sec. 6.1. *Hint:* The support at  $A$  resists only a horizontal force and a bending moment.

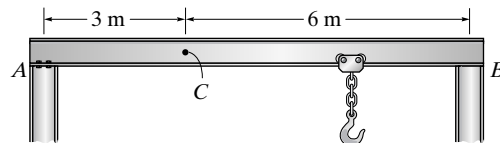
**\*6-8.** Solve Prob. 6-7 using the Müller-Breslau principle.



**Probs. 6-7/8**

**6-9.** Draw the influence line for (a) the vertical reaction at  $A$ , (b) the shear at  $C$ , and (c) the moment at  $C$ . Solve this problem using the basic method of Sec. 6.1. Assume  $A$  is a roller and  $B$  is a pin.

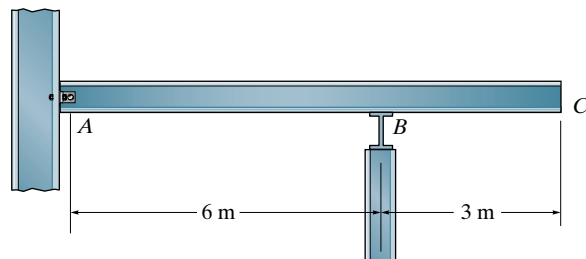
**6-10.** Solve Prob. 6-9 using Müller-Breslau's principle.



**Probs. 6-9/10**

**6-11.** Draw the influence lines for (a) the vertical reaction at  $B$ , (b) the shear just to the left of  $B$ , and (c) the moment at  $B$ . Solve this problem using the basic method of Sec. 6.1. Assume  $A$  is a pin and  $B$  is a roller.

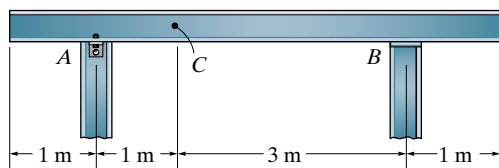
**\*6-12.** Solve Prob. 6-11 using Müller-Breslau's principle.



**Probs. 6-11/12**

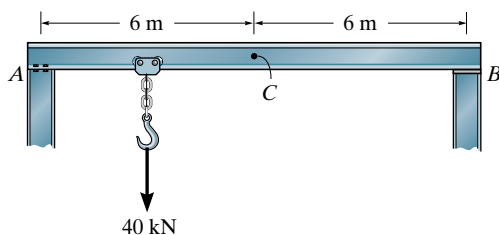
**6-13.** Draw the influence line for (a) the moment at  $C$ , (b) the shear just to the right of the support at  $B$ , and (c) the vertical reaction at  $B$ . Solve this problem using the basic method of Sec. 6.1. Assume  $A$  is a pin and  $B$  is a roller.

**6-14.** Solve Prob. 6-13 using the Müller-Breslau principle.



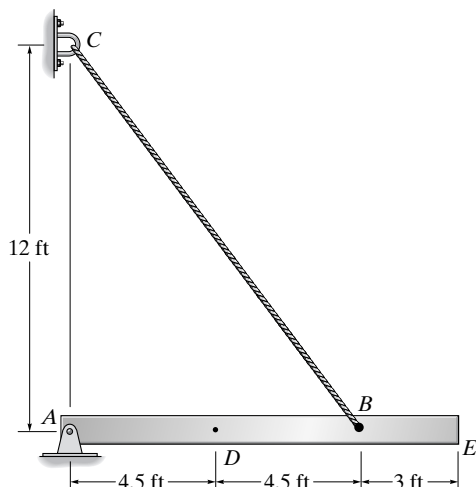
**Probs. 6-13/14**

**6-15.** The beam is subjected to a uniform dead load of  $1.2 \text{ kN/m}$  and a single live load of  $40 \text{ kN}$ . Determine (a) the maximum moment created by these loads at  $C$ , and (b) the maximum positive shear at  $C$ . Assume  $A$  is a pin, and  $B$  is a roller.



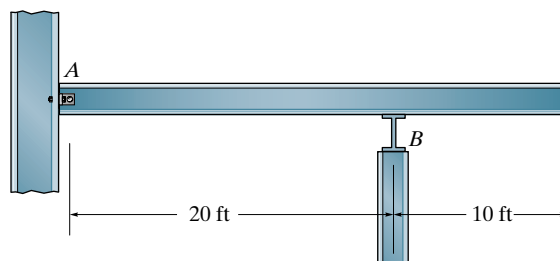
**Prob. 6-15**

**\*6-16.** Draw the influence line for (a) the force in the cable  $BC$ , (b) the vertical reaction at  $A$ , and (c) the moment at  $D$ .



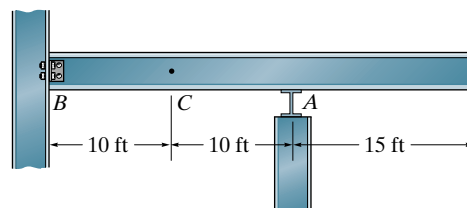
**Prob. 6-16**

**6-17.** A uniform live load of  $300 \text{ lb/ft}$  and a single live concentrated force of  $1500 \text{ lb}$  are to be placed on the beam. The beam has a weight of  $150 \text{ lb/ft}$ . Determine (a) the maximum vertical reaction at support  $B$ , and (b) the maximum negative moment at point  $B$ . Assume the support at  $A$  is a pin and  $B$  is a roller.



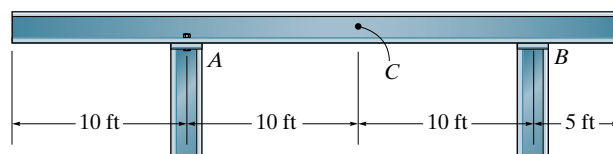
**Prob. 6-17**

**6-18.** The beam supports a uniform dead load of  $0.4 \text{ k/ft}$ , a live load of  $1.5 \text{ k/ft}$ , and a single live concentrated force of  $8 \text{ k}$ . Determine (a) the maximum positive moment at  $C$ , and (b) the maximum positive vertical reaction at  $B$ . Assume  $A$  is a roller and  $B$  is a pin.



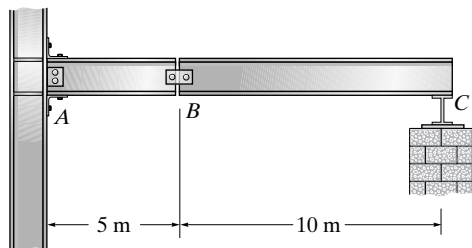
**Prob. 6-18**

**6-19.** The beam is used to support a dead load of  $0.6 \text{ k/ft}$ , a live load of  $2 \text{ k/ft}$  and a concentrated live load of  $8 \text{ k}$ . Determine (a) the maximum positive (upward) reaction at  $A$ , (b) the maximum positive moment at  $C$ , and (c) the maximum positive shear just to the right of the support at  $A$ . Assume the support at  $A$  is a pin and  $B$  is a roller.



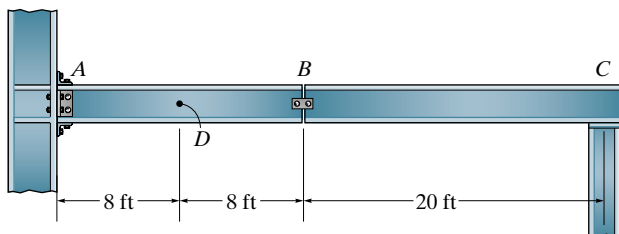
**Prob. 6-19**

**\*6-20.** The compound beam is subjected to a uniform dead load of  $1.5 \text{ kN/m}$  and a single live load of  $10 \text{ kN}$ . Determine (a) the maximum negative moment created by these loads at  $A$ , and (b) the maximum positive shear at  $B$ . Assume  $A$  is a fixed support,  $B$  is a pin, and  $C$  is a roller.



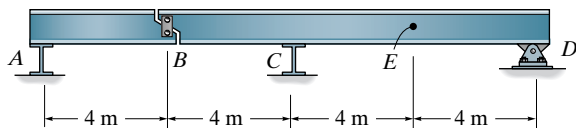
**Prob. 6-20**

**6-21.** Where should a single  $500\text{-lb}$  live load be placed on the beam so it causes the largest moment at  $D$ ? What is this moment? Assume the support at  $A$  is fixed,  $B$  is pinned, and  $C$  is a roller.



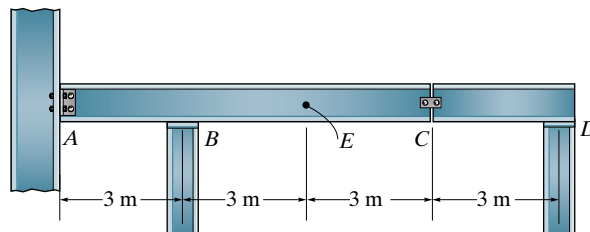
**Prob. 6-21**

**6-22.** The beam is subjected to a uniform live load of  $1.2 \text{ kN/m}$ , a dead load of  $0.5 \text{ kN/m}$ , and a single live load of  $40 \text{ kN}$ . Determine (a) the maximum positive moment created by these loads at  $E$ , and (b) the maximum positive shear at  $E$ . Assume  $A$  and  $C$  are roller, and  $B$  is a short link.



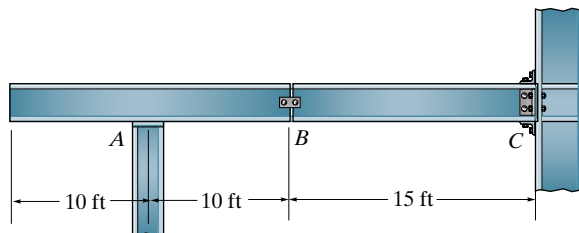
**Prob. 6-22**

**6-23.** The beam supports a uniform dead load of  $500 \text{ N/m}$  and a single live concentrated force of  $3000 \text{ N}$ . Determine (a) the maximum negative moment at  $E$ , and (b) the maximum positive shear at  $E$ . Assume the support at  $A$  is a pin,  $B$  and  $D$  are rollers, and  $C$  is a pin.



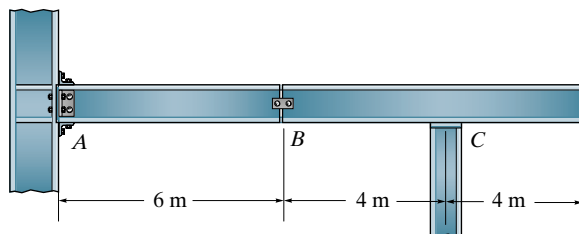
**Prob. 6-23**

**\*6-24.** The beam is used to support a dead load of  $400 \text{ lb/ft}$ , a live load of  $2 \text{ k/ft}$ , and a concentrated live load of  $8 \text{ k}$ . Determine (a) the maximum positive vertical reaction at  $A$ , (b) the maximum positive shear just to the right of the support at  $A$ , and (c) the maximum negative moment at  $C$ . Assume  $A$  is a roller,  $C$  is fixed, and  $B$  is pinned.



**Prob. 6-24**

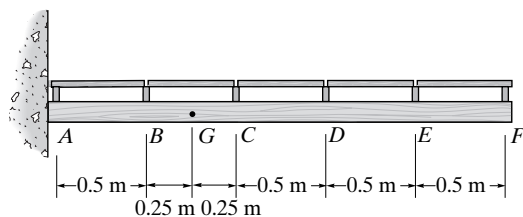
**6-25.** Draw the influence line for (a) the vertical reaction at  $A$ , (b) the shear just to the right of  $B$ , and (c) the moment at  $A$ . Assume  $A$  is fixed,  $C$  is a roller, and  $B$  is a pin.



**Prob. 6-25**

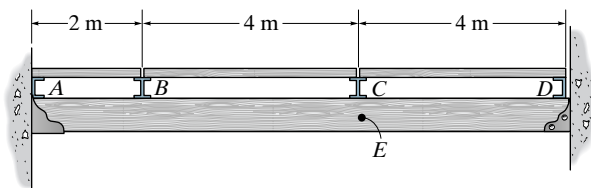
## Sec. 6.4

**6-26.** A uniform live load of  $1.8 \text{ kN/m}$  and a single concentrated live force of  $4 \text{ kN}$  are placed on the floor beams. Determine (a) the maximum positive shear in panel  $BC$  of the girder and (b) the maximum moment in the girder at  $G$ .



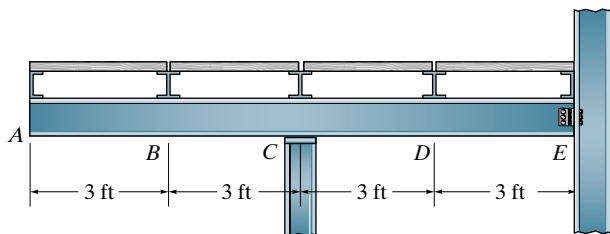
**Prob. 6-26**

**6-27.** Draw the influence line for the moment at  $E$  in the girder. Determine the maximum positive moment in the girder at  $E$  if a single concentrated live force of  $5 \text{ kN}$  and a uniform live load of  $1.5 \text{ kN/m}$  can be placed on the floor beams. Assume  $A$  is a pin and  $D$  is a roller.



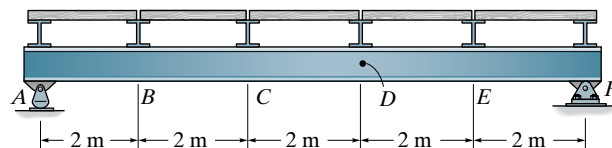
**Prob. 6-27**

**\*6-28.** A uniform live load of  $2 \text{ k/ft}$  and a single concentrated live force of  $6 \text{ k}$  are placed on the floor beams. If the beams also support a uniform dead load of  $350 \text{ lb/ft}$ , determine (a) the maximum positive shear in panel  $CD$  of the girder and (b) the maximum negative moment in the girder at  $D$ . Assume the support at  $C$  is a roller and  $E$  is a pin.



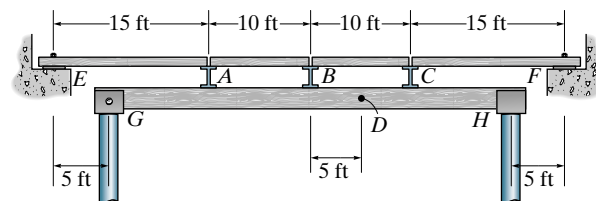
**Prob. 6-28**

**6-29.** Draw the influence line for (a) the shear in panel  $BC$  of the girder, and (b) the moment at  $D$ .



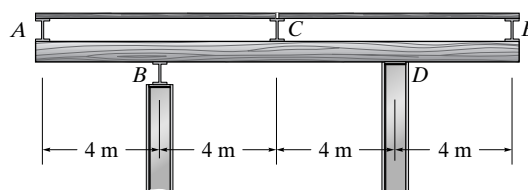
**Prob. 6-29**

**6-30.** A uniform live load of  $250 \text{ lb/ft}$  and a single concentrated live force of  $1.5 \text{ k}$  are to be placed on the floor beams. Determine (a) the maximum positive shear in panel  $AB$ , and (b) the maximum moment at  $D$ . Assume only vertical reaction occur at the supports.



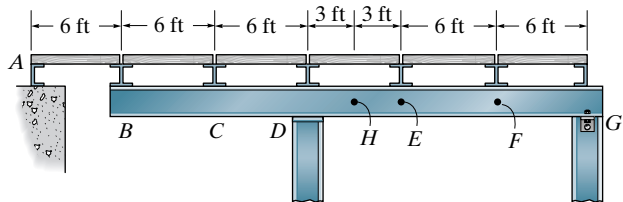
**Prob. 6-30**

**6-31.** A uniform live load of  $16 \text{ kN/m}$  and a single concentrated live force of  $34 \text{ kN}$  are placed on the top beams. If the beams also support a uniform dead load of  $3 \text{ kN/m}$ , determine (a) the maximum positive shear in panel  $BC$  of the girder and (b) the maximum positive moment in the girder at  $C$ . Assume  $B$  is a roller and  $D$  is a pin.



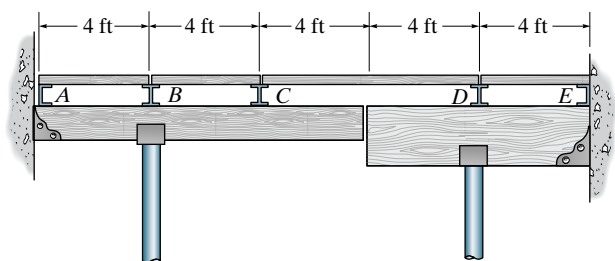
**Prob. 6-31**

**\*6-32.** A uniform live load of 0.4 k/ft and a single concentrated live force of 6 k are placed on the floor beams. Determine (a) the maximum positive shear in panel *EF* of the girder, and (b) the maximum positive moment at *H*.



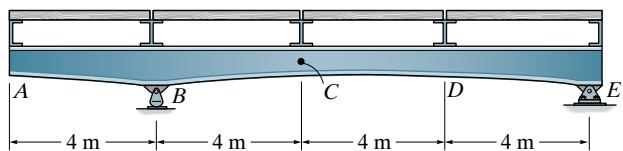
**Prob. 6-32**

**6-33.** A uniform live load of 300 lb/ft and a single concentrated live force of 2 k are to be placed on the floor beams. Determine (a) the maximum negative shear in panel *AB*, and (b) the maximum negative moment at *B*. Assume the supports at *A* and *E* are pins and the pipe columns only exert vertical reactions on the beams.



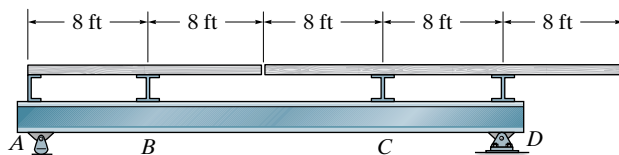
**Prob. 6-33**

**6-34.** A uniform live load of 8 kN/m and a single concentrated live force of 12 kN are placed on the floor beams. If the beams also support a uniform dead load of 400 N/m, determine (a) the maximum positive shear in panel *CD* of the girder and (b) the maximum positive moment in the girder at *C*.



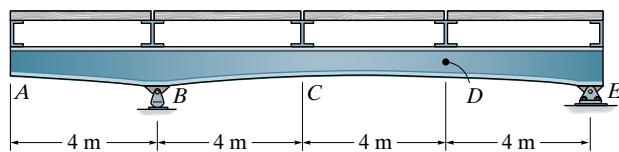
**Prob. 6-34**

**6-35.** Draw the influence line for the shear in panel *CD* of the girder. Determine the maximum negative live shear in panel *CD* due to a uniform live load of 500 lb/ft acting on the top beams.



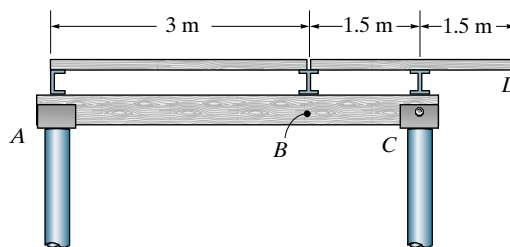
**Prob. 6-35**

**\*6-36.** A uniform live load of 6.5 kN/m and a single concentrated live force of 15 kN are placed on the floor beams. If the beams also support a uniform dead load of 600 N/m, determine (a) the maximum positive shear in panel *CD* of the girder and (b) the maximum positive moment in the girder at *D*.



**Prob. 6-36**

**6-37.** A uniform live load of 1.75 kN/m and a single concentrated live force of 8 kN are placed on the floor beams. If the beams also support a uniform dead load of 250 N/m, determine (a) the maximum negative shear in panel *BC* of the girder and (b) the maximum positive moment at *B*.

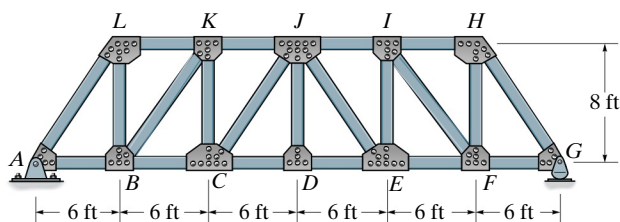


**Prob. 6-37**

## Sec. 6.5

**6-38.** Draw the influence line for the force in (a) member  $KJ$  and (b) member  $CJ$ .

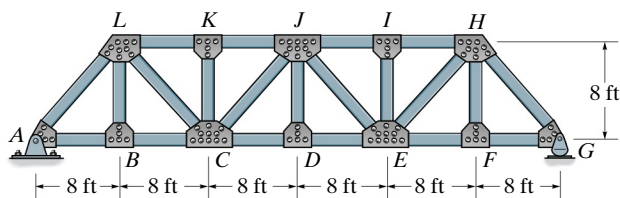
**6-39.** Draw the influence line for the force in (a) member  $JI$ , (b) member  $IE$ , and (c) member  $EF$ .



**Probs. 6-38/39**

**\*6-40.** Draw the influence line for the force in member  $KJ$ .

**6-41.** Draw the influence line for the force in member  $JE$ .

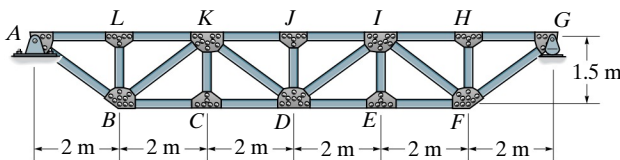


**Probs. 6-40/41**

**6-42.** Draw the influence line for the force in member  $CD$ .

**6-43.** Draw the influence line for the force in member  $JK$ .

**\*6-44.** Draw the influence line for the force in member  $DK$ .

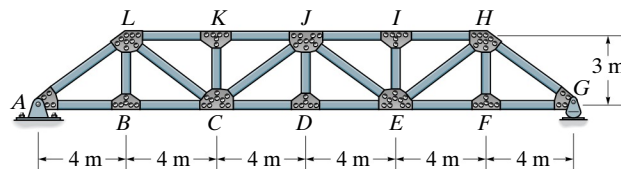


**Probs. 6-42/43/44**

**6-45.** Draw the influence line for the force in (a) member  $EH$  and (b) member  $JE$ .

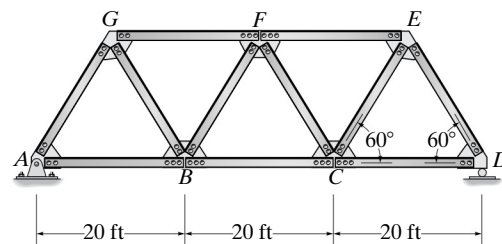
**6-46.** Draw the influence line for the force in member  $JI$ .

**6-47.** Draw the influence line for the force in member  $AL$ .



**Probs. 6-45/46/47**

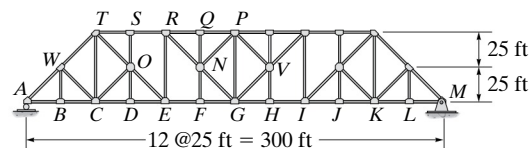
**\*6-48.** Draw the influence line for the force in member  $BC$  of the Warren truss. Indicate numerical values for the peaks. All members have the same length.



**Prob. 6-48**

**6-49.** Draw the influence line for the force in member  $CD$  of the Baltimore truss.

**6-50.** Draw the influence line for the force in member  $PG$  of the Baltimore truss.



**Probs. 6-49/50**

**6-51.** Draw the influence line for the force in member  $RQ$  of the Baltimore truss.

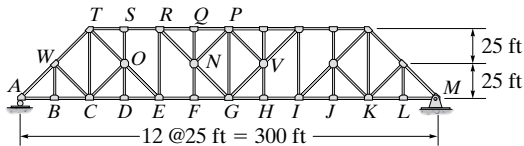
**\*6-52.** Draw the influence line for the force in member  $TC$  of the Baltimore truss.

**6-53.** Draw the influence line for the force in member  $NP$  of the Baltimore truss.

**6-54.** Draw the influence line for the force in member  $RN$  of the Baltimore truss.

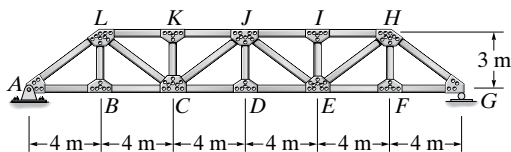
**6-55.** Draw the influence line for the force in member  $NG$  of the Baltimore truss.

**\*6-56.** Draw the influence line for the force in member  $CO$  of the Baltimore truss.



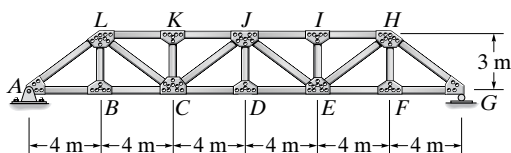
**Probs. 6-51/52/53/54/55/56**

**6-57.** Draw the influence line for the force in member  $CD$ .



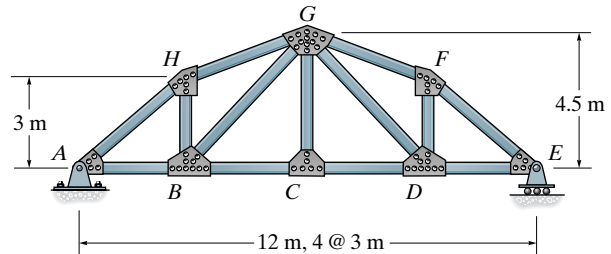
**Prob. 6-57**

**6-58.** Draw the influence line for the force in member  $KJ$ .



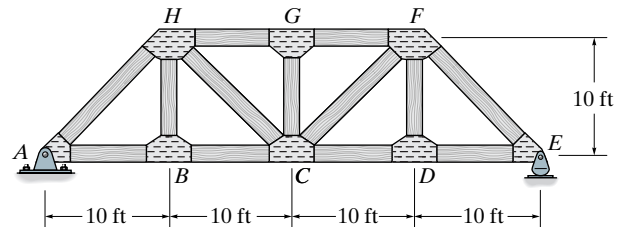
**Prob. 6-58**

**6-59.** Draw the influence line for the force in member  $GD$ , then determine the maximum force (tension or compression) that can be developed in this member due to a uniform live load of  $3 \text{ kN/m}$  that acts on the bridge deck along the bottom cord of the truss.



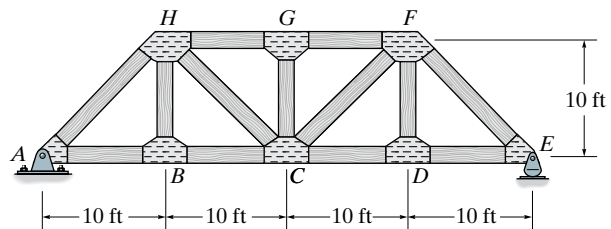
**Prob. 6-59**

**\*6-60.** Draw the influence line for the force in member  $CD$ , and then determine the maximum force (tension or compression) that can be developed in this member due to a uniform live load of  $800 \text{ lb/ft}$  which acts along the bottom cord of the truss.



**Prob. 6-60**

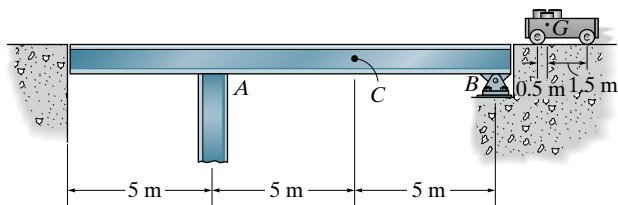
**6-61.** Draw the influence line for the force in member  $CF$ , and then determine the maximum force (tension or compression) that can be developed in this member due to a uniform live load of  $800 \text{ lb/ft}$  which is transmitted to the truss along the bottom cord.



**Prob. 6-61**

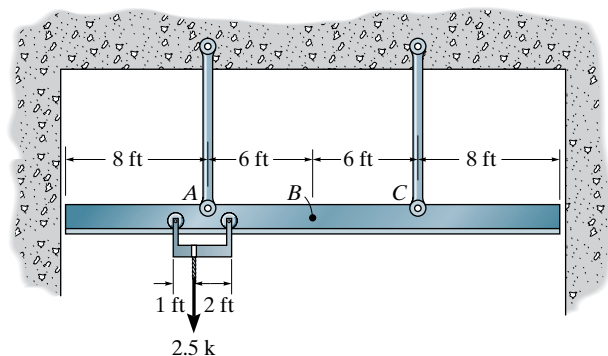
## Sec. 6.6

**6-62.** Determine the maximum moment at point  $C$  on the single girder caused by the moving dolly that has a mass of  $2\text{ Mg}$  and a mass center at  $G$ . Assume  $A$  is a roller.



**Prob. 6-62**

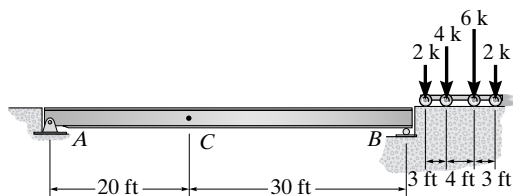
**6-63.** Determine the maximum moment in the suspended rail at point  $B$  if the rail supports the load of  $2.5\text{ k}$  on the trolley.



**Prob. 6-63**

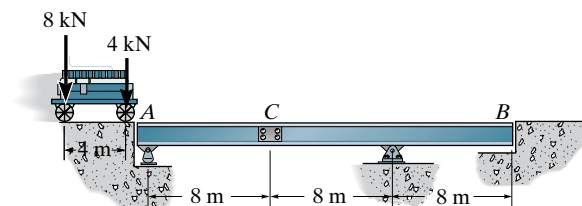
**\*6-64.** Determine the maximum live moment at  $C$  caused by the moving loads.

**6-65.** Determine the maximum live shear at  $C$  caused by the moving loads.



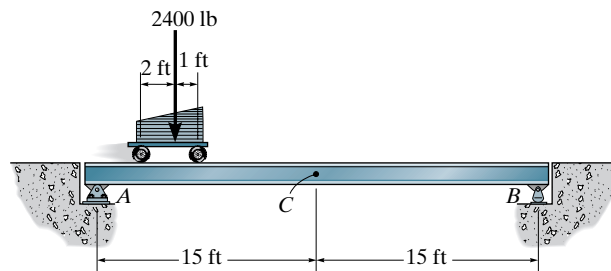
**Probs. 6-64/65**

**6-66.** Determine the maximum positive moment at the splice  $C$  on the side girder caused by the moving load which travels along the center of the bridge.



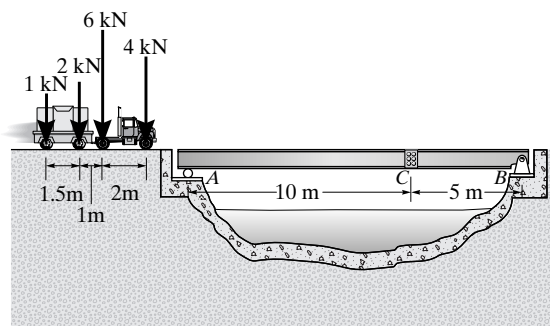
**Prob. 6-66**

**6-67.** Determine the maximum moment at  $C$  caused by the moving load.



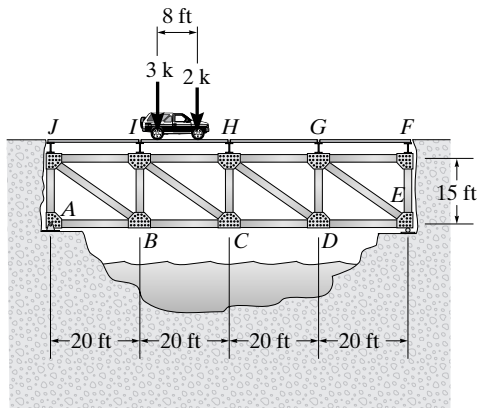
**Prob. 6-67**

**\*6-68.** The truck and trailer exerts the wheel reactions shown on the deck of the girder bridge. Determine the largest moment it exerts in the splice at  $C$ . Assume the truck travels in *either direction* along the *center* of the deck, and therefore transfers *half* of the load shown to each of the two side girders. Assume the splice is a fixed connection and, like the girder, can support both shear and moment.



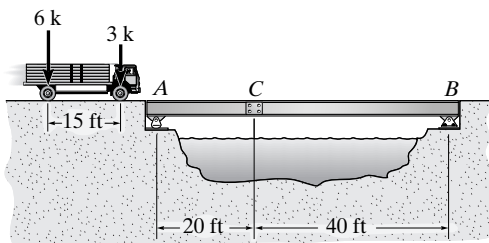
**Prob. 6-68**

**6-69.** Draw the influence line for the force in member  $GF$  of the bridge truss. Determine the maximum live force (tension or compression) that can be developed in the member due to a 5-k truck having the wheel loads shown. Assume the truck can travel in *either direction* along the *center of the deck*, so that *half* the load shown is transferred to each of the two side trusses. Also assume the members are pin connected at the gusset plates.



**Prob. 6-69**

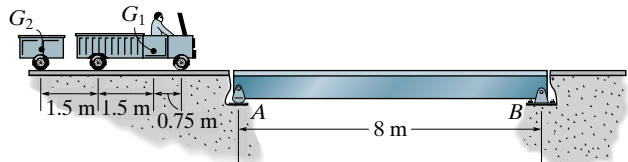
**6-70.** The 9-k truck exerts the wheel reactions shown on the deck of a girder bridge. Determine (a) the largest live shear it creates in the splice at  $C$ , and (b) the largest moment it exerts in the splice. Assume the truck travels in *either direction* along the *center of the deck*, and therefore transfers *half* of the load shown in each of the two side girders. Assume the splice is a fixed connection and, like the girder, can support both shear and moment.



**Prob. 6-70**

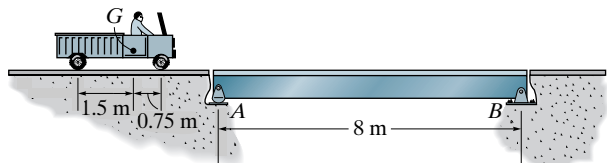
## Sec. 6.7

**6-71.** The truck has a mass of 4 Mg and mass center at  $G_1$ , and the trailer has a mass of 1 Mg and mass center at  $G_2$ . Determine the absolute maximum live moment developed in the bridge.



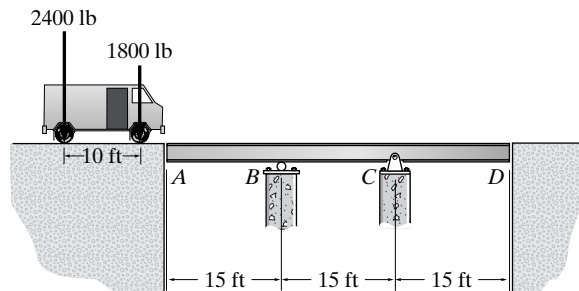
**Prob. 6-71**

**\*6-72.** The truck has a mass of 4-Mg and mass center at  $G$ . Determine the absolute maximum live moment developed in the bridge.



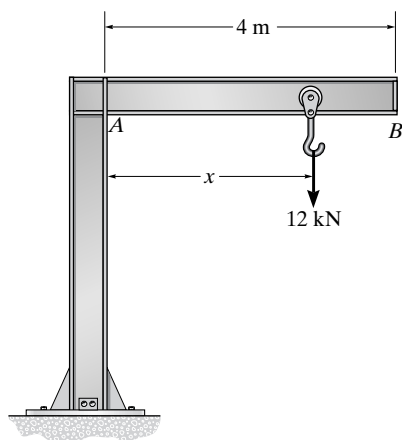
**Prob. 6-72**

**6-73.** Determine the absolute maximum live moment in the girder bridge due to the loading shown. The load is applied directly to the girder.



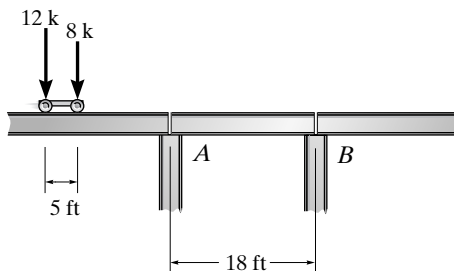
**Prob. 6-73**

**6-74.** Determine the absolute maximum live shear and absolute maximum live moment in the jib beam  $AB$  due to the crane loading. The end constraints require  $0.1 \text{ m} \leq x \leq 3.9 \text{ m}$ .



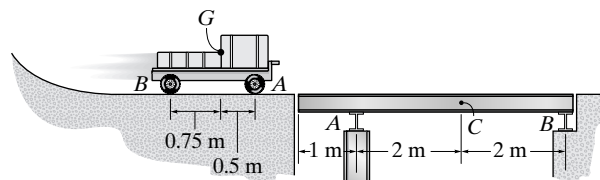
**Prob. 6-74**

**6-75.** The maximum wheel loadings for the wheels of a crane that is used in an industrial building are given. The crane travels along the runway girders that are simply supported on columns. Determine (a) the absolute maximum shear in an intermediate girder  $AB$ , and (b) the absolute maximum moment in the girder.



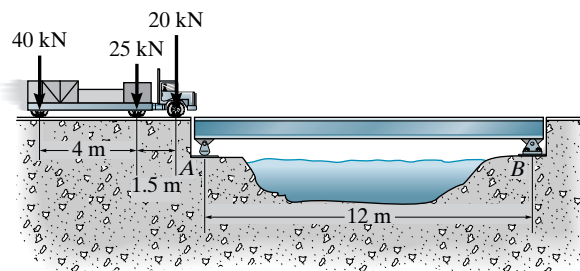
**Prob. 6-75**

**\*6-76.** The cart has a mass of 2 Mg and center of mass at  $G$ . Determine the maximum live moment created in the side girder at  $C$  as it crosses the bridge. Assume the cart travels along the *center* of the deck, so that *half* the load shown is transferred to each of the two side girders.



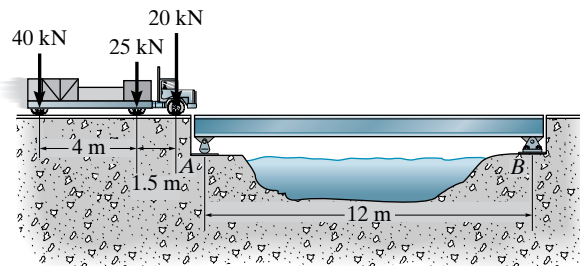
**Prob. 6-76**

**6-77.** Determine the absolute maximum shear in the beam due to the loading shown.



**Prob. 6-77**

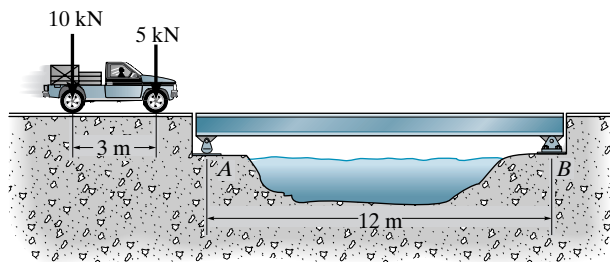
**6-78.** Determine the absolute maximum moment in the beam due to the loading shown.



**Prob. 6-78**

**6-79.** Determine the absolute maximum shear in the bridge girder due to the loading shown.

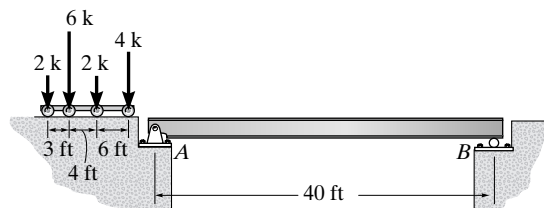
**\*6-80.** Determine the absolute maximum moment in the bridge girder due to the loading shown.



**Probs. 6-79/80**

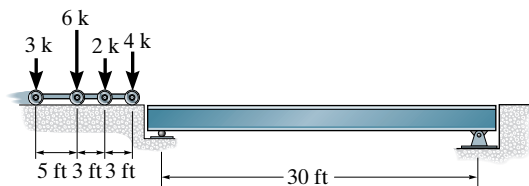
6

**6-81.** Determine the absolute maximum live moment in the bridge.



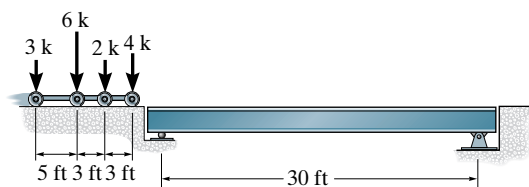
**Prob. 6-81**

**6-82.** Determine the absolute maximum shear in the beam due to the loading shown.



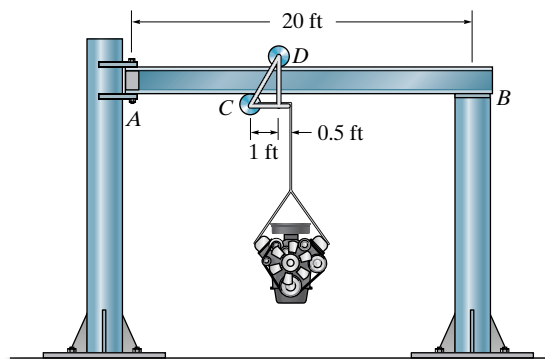
**Prob. 6-82**

**6-83.** Determine the absolute maximum moment in the bridge due to the loading shown.



**Prob. 6-83**

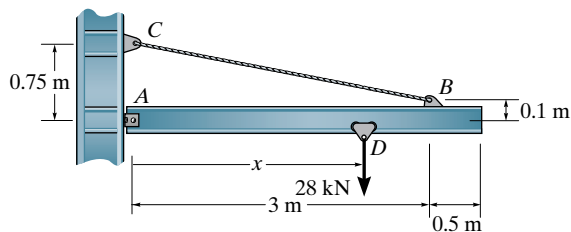
**\*6-84.** The trolley rolls at C and D along the bottom and top flange of beam AB. Determine the absolute maximum moment developed in the beam if the load supported by the trolley is 2 k. Assume the support at A is a pin and at B a roller.



**Prob. 6-84**

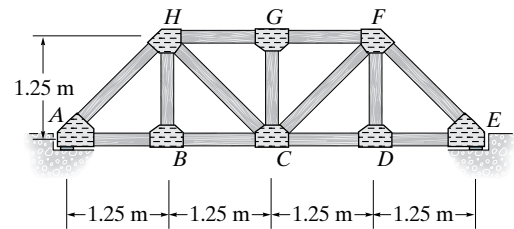
## PROJECT PROBLEMS

**6-1P.** The chain hoist on the wall crane can be placed anywhere along the boom ( $0.1 \text{ m} < x < 3.4 \text{ m}$ ) and has a rated capacity of 28 kN. Use an impact factor of 0.3 and determine the absolute maximum bending moment in the boom and the maximum force developed in the tie rod  $BC$ . The boom is pinned to the wall column at its left end  $A$ . Neglect the size of the trolley at  $D$ .

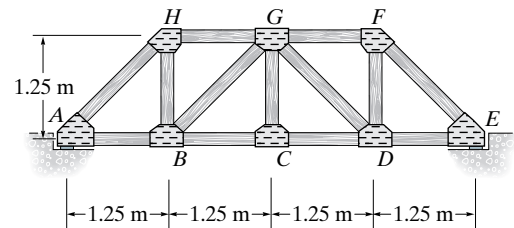


**Prob. 6-1P**

**6-2P.** A simply supported pedestrian bridge is to be constructed in a city park and two designs have been proposed as shown in case *a* and case *b*. The truss members are to be made from timber. The deck consists of 1.5-m-long planks that have a mass of  $20 \text{ kg/m}^2$ . A local code states the live load on the deck is required to be  $5 \text{ kPa}$  with an impact factor of 0.2. Consider the deck to be simply supported on stringers. Floor beams then transmit the load to the bottom joints of the truss. (See Fig. 6-23.) In each case find the member subjected to the largest tension and largest compression load and suggest why you would choose one design over the other. Neglect the weights of the truss members.



case *a*



case *b*

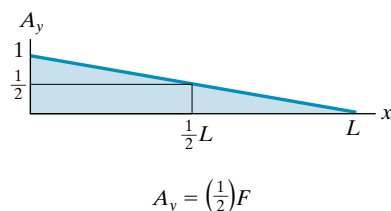
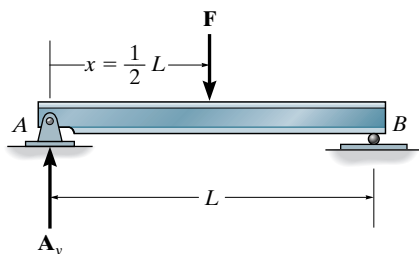
**Prob. 6-2P**

## CHAPTER REVIEW

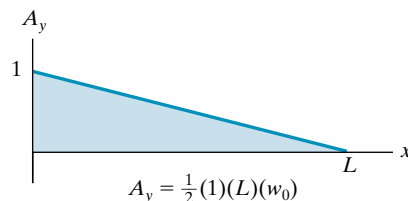
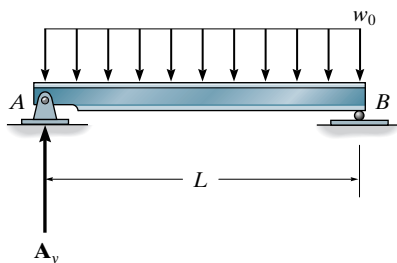
An influence line indicates the value of a reaction, shear, or moment at a specific point on a member, as a unit load moves over the member.

Once the influence line for a reaction, shear, or moment (function) is constructed, then it will be possible to locate the live load on the member to produce the maximum positive or negative value of the function.

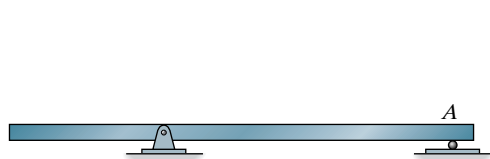
A concentrated live force is applied at the positive (negative) peaks of the influence line. The value of the function is then equal to the product of the influence line ordinate and the magnitude of the force.



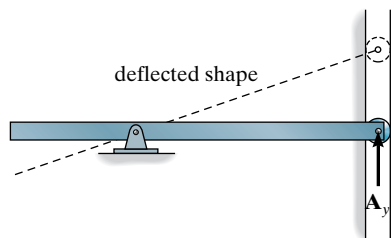
A uniform distributed load extends over a positive (negative) region of the influence line. The value of the function is then equal to the product of the area under the influence line for the region and the magnitude of the uniform load.



The general shape of the influence line can be established using the Müller-Breslau principle, which states that the influence line for a reaction, shear, or moment is to the same scale as the deflected shape of the member when it is acted upon by the reaction, shear, or moment.



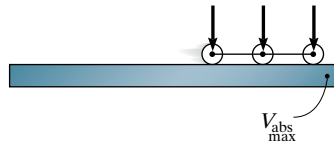
(a)



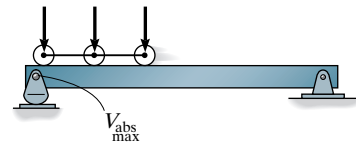
(b)

Influence lines for floor girders and trusses can be established by placing the unit load at each panel point or joint, and calculating the value of the required reaction, shear, or moment.

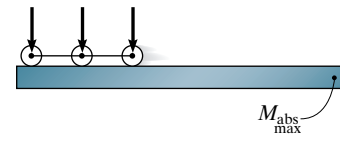
When a series of concentrated loads pass over the member, then the various positions of the load on the member have to be considered to determine the largest shear or moment in the member. In general, place the loadings so that each contributes its maximum influence, as determined by multiplying each load by the ordinate of the influence line. This process of finding the actual position can be done using a trial-and-error procedure, or by finding the change in either the shear or moment when the loads are moved from one position to another. Each moment is investigated until a negative change in the shear or moment occurs. Once this happens the previous position will define the critical loading.



Absolute maximum *shear* in a cantilever or simply supported beam will occur at a support, when one of the loads is placed next to the support.



Absolute maximum *moment* in a cantilevered beam occurs when the series of concentrated loads are placed at the farthest point away from the fixed support.



To determine the absolute maximum moment in a simply supported beam, the resultant of the force system is first determined. Then it, along with one of the concentrated forces in the system is positioned so that these two forces are equidistant from the centerline of the beam. The maximum moment then occurs under the selected force. Each force in the system is selected in this manner, and by comparison the largest for all these cases is the absolute maximum moment.(f)

